The Boundary Test Program

Applying $Q = F \cdot \beta \cdot d^2 \cdot Z$ to Coherent Pattern Boundaries Across Scales

Registry: [HOWL-PHYS-4-2026]

Series Path: [HOWL-PHYS-1-2026] → [HOWL-MATH-1-2026] → [HOWL-PHYS-3-2026] → [HOWL-PHYS-4-2026]

DOI: 10.5281/zenodo.19665995

Date: March 2026

Domain: Foundational Physics / Measurement Theory

Status: Complete

AI Usage Disclosure: Only the top metadata, figures, refs and final copyright sections were edited by the author. All paper content was LLM-generated using Anthropic's Claude 4.5 Sonnet.

I. ABSTRACT

This paper defines a testing program that connects three prior results: the coherent pattern boundaries and anomaly correlations of [HOWL-PHYS-1-2026], the geometric invariant $\beta = \pi/4$ and unified equation $Q = F \cdot \beta \cdot d^2 \cdot Z$ of [HOWL-MATH-1-2026], and the experimental gap in G measurements across Earth's Hill sphere documented in [HOWL-PHYS-3-2026].

The program asks one question: does the $Q = F \cdot \beta \cdot d^2 \cdot Z$ framework, proven for geometric cross-sections across nine domains, apply to the coherent pattern boundaries where measurement anomalies have been documented?

The paper does not answer this question. It classifies which boundaries are geometric and which are not. It specifies seven tests ordered from immediately achievable to far future. It identifies three tests that are reanalyses of existing published data requiring no new experiments. It commits to a kill switch: if all three existing-data tests produce null results, the remaining tests are not motivated and the program stops.

One test — the Hubble tension cumulative transit calculation — is identified as underspecified. The per-boundary transit correction magnitude is not provided by the prior papers and has not been derived. This missing piece is stated as a primary limitation. The test cannot be executed until the per-transit correction is established by theoretical work this paper does not attempt.

Every premise is from the institution's own published literature or from the prior papers in this series. No new physics is proposed. No new mathematics is introduced. The

contribution is the specification of a test program connecting established results that have not been connected.

II. FOUNDATIONS

2.1 What MATH-1 Proved

[HOWL-MATH-1-2026] proved that nine equations across nine domains of applied mathematics share a common structure:

$$\mathbf{Q = F \cdot \beta \cdot d^2 \cdot Z}$$

where $\beta = \pi/4$ is the ratio of circular cross-sectional area to rectilinear bounding area. The proof is algebraic identity — each equation is algebraically identical to its standard form. The nine domains are geometry, pipe flow, drag, orifice flow, capacitance, electromagnetic aperture, antenna theory, beam optics, and thermal radiation. Every instance involves a circular or elliptical geometry interfacing with a rectilinear measurement frame. The geometric factor $\beta \cdot d^2$ is invariant across all nine domains. F (driving term) and Z (domain-specific impedance) vary.

The scope of the proof: geometric cross-sections. Circular and elliptical areas interfacing with rectilinear measurement. The proof has not been shown to apply to non-geometric boundaries.

2.2 What PHYS-1 Identified

[HOWL-PHYS-1-2026] identified coherent pattern boundaries at multiple scales — electron vacuum polarization clouds, hadron confinement boundaries, proton internal structure, atomic shells, planetary gravitational wells, stellar boundaries, galactic structure, cosmological structure. Interior and exterior readings differ across these boundaries. Three measurement anomalies — the Hubble tension, the proton radius puzzle, and the muon g-2 discrepancy — correlate with boundary transit count or interaction depth.

The scope of the identification: qualitative. The anomaly correlation is a pattern across three data points. No mathematical framework was applied. The correlation warrants investigation but does not constitute proof.

2.3 What PHYS-3 Established

[HOWL-PHYS-3-2026] established that every direct measurement of the gravitational constant G in the 227-year experimental record has been performed inside Earth's Hill sphere — at 0.45% or less of the Hill sphere distance. The persistent disagreement between G measurements from different experimental groups has not been resolved by two centuries of systematic error investigation. PHYS-3 proposed a direct G measurement at the L1 or L2 Lagrange point as the first boundary-crossing test. PHYS-3 also documented that every coupling whose boundary has been identified and whose running has been measured (α, αs) shows a consistent single curve with no method disagreement, while G — the one coupling whose boundary has not been identified — shows persistent method disagreement.

The scope of the establishment: the experimental gap is factual. The interpretation — that G may be boundary-dependent — is a hypothesis awaiting the proposed test.

2.4 What Has Not Been Connected

MATH-1 provides a framework proven for geometric cross-sections. PHYS-1 catalogs boundaries where anomalies correlate with boundary structure. PHYS-3 identifies a specific untested boundary for a specific measurement. These three results have not been connected. Whether the geometric framework applies to the physical boundaries is an open question. The answer might be no. This paper defines the tests that would determine the answer.

III. WHICH BOUNDARIES ARE GEOMETRIC?

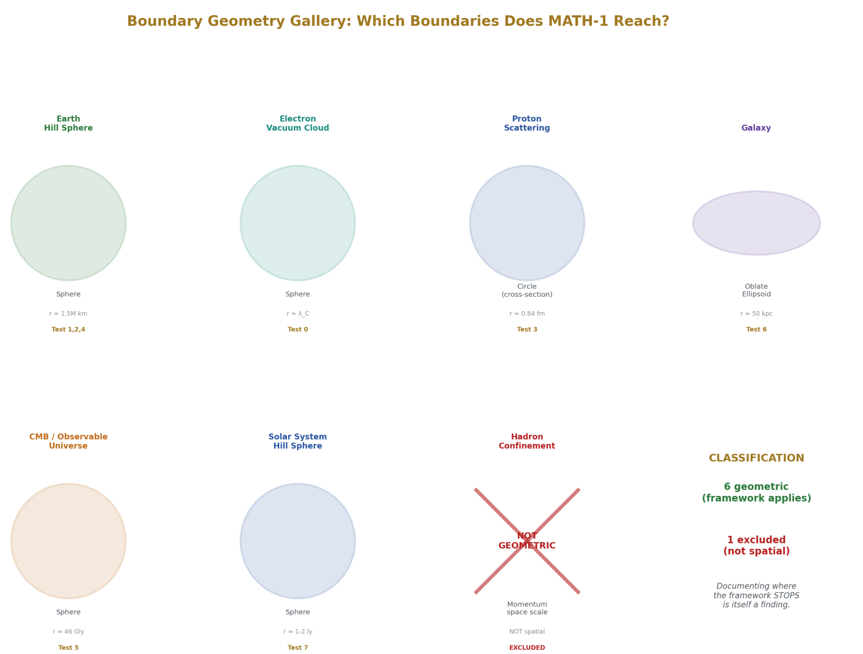

Figure 1: Fig. 1: Six boundaries classified as geometric (sphere, ellipse, scattering cross-section) and one excluded (confinement — not spatial). The framework's scope is visible.

3.1 The Criterion

MATH-1's framework applies to boundaries that are spatial geometries — circles, spheres, or ellipses — interfacing with a rectilinear measurement frame. Not all boundaries cataloged in PHYS-1 meet this criterion. Each boundary is evaluated individually below.

3.2 Earth's Hill Sphere

The Hill sphere is a sphere. It has a calculable radius (~1.5 million km). Its cross-section is circular. A measurement probe approaching Earth from outside encounters a spherical boundary between the solar reference frame — in which Earth is a point mass — and Earth's coherent gravitational interior where Earth's structure dominates.

The geometry is spatial and circular. The interface is between an interior frame and an exterior frame. The MATH-1 framework requires exactly this: a circular geometry interfacing with a rectilinear measurement frame.

Classification: geometric. Framework applies in principle.

3.3 Solar System Hill Sphere

Same structure as Earth's Hill sphere at a larger scale. The solar system's Hill sphere is a sphere with radius ~1-2 light years. The interface is between the Sun's coherent gravitational domain and the galactic field. Spatial, spherical, circular cross-section.

Classification: geometric. Framework applies in principle. Untestable with current technology — no instruments at this boundary.

3.4 Proton

The proton is a quantum object whose charge radius is defined operationally through scattering cross-sections. The electron-proton and muon-proton scattering cross-sections are circular — they are literally π r^2 terms at each probe energy, measured in the center-of-mass frame. The probe interacts with the proton at a characteristic radius determined by the probe's energy and mass. The cross-section at each radius is circular.

The proton is not a classical sphere. But the measurement of the proton — the scattering cross-section — is a circular geometric cross-section in the precise sense MATH-1 requires. The framework applies to the measurement geometry, not to the proton's internal ontology.

Classification: geometric at the level of measurement cross-sections. Framework applies in principle. The proton radius puzzle is the test case.

3.5 Electron Vacuum Polarization Cloud

The virtual particle cloud surrounding a bare electron charge is spherically symmetric. The boundary between bare charge ($\alpha \approx 1/127$ at high energy) and screened charge (α

$\approx 1/137$ at low energy) is a sphere at the characteristic Compton wavelength scale of the electron (~3.86×10^{-13} m). The measurement of α at different probe energies involves scattering cross-sections — circular geometric cross-sections at each energy.

Classification: geometric at the level of scattering cross-sections. The running of α is the established measurement across this boundary. This is the calibration case — the known result the framework must reproduce before being applied to unknown cases.

3.6 Hadron Confinement Boundary

Confinement is a property of QCD. The boundary where asymptotic freedom gives way to confinement is defined in momentum space — it is a scale in the running coupling constant, not a geometric surface in position space. There is no spatial sphere. There is no circular cross-section interfacing with a rectilinear frame in the sense MATH-1 requires.

Classification: not geometric in the MATH-1 sense. The framework does not apply directly. This boundary is outside scope.

This is not a failure. It is a scope finding. The framework applies to spatial geometric boundaries. Confinement is not a spatial geometric boundary. Documenting where the framework stops is as important as documenting where it applies.

3.7 Galaxy

The galactic disk is not spherical. It is oblate — roughly elliptical in cross-section when viewed edge-on. MATH-1 Section III establishes that β applies to all ellipses: the area of any ellipse equals β times its bounding rectangle, at all eccentricities.

The galactic boundary — where the galaxy's coherent gravitational structure gives way to the intergalactic field — is approximately elliptical.

Classification: geometric with the ellipse generalization. Framework applies in principle. Calculationally difficult — the galactic boundary is not a clean ellipse and the rectilinear measurement frame is not obvious at galactic scales.

3.8 Observable Universe / CMB Boundary

The observable universe is approximately spherical. The cosmic microwave background is emitted from the surface of last scattering — a spherical surface. Light transiting from the CMB to our instruments crosses this spherical boundary plus every coherent gravitational structure between the CMB surface and our instruments.

Classification: geometric. The Hubble tension is the test case. The most complex and currently underspecified calculation in the program.

3.9 Summary

Boundary	Geometry	Spatial?	Classification	Test Case
Earth Hill sphere	Sphere	Yes	Geometric	G measurement
Solar system Hill sphere	Sphere	Yes	Geometric	Not testable now
Proton	Measurement cross-section is circular	Yes	Geometric	Proton radius puzzle
Electron vacuum cloud	Sphere / scattering cross-section	Yes	Geometric	Running of α (calibration)
Hadron confinement	Momentum-space scale	No	Not geometric — outside scope	—
Galaxy	Oblate ellipsoid	Yes	Geometric (ellipse)	Rotation curves
Observable universe / CMB	Sphere	Yes	Geometric	Hubble tension

Six of seven boundaries are geometric. One is not. The framework's scope covers six boundaries. The exclusion is documented.

IV. CALIBRATION: THE RUNNING OF α

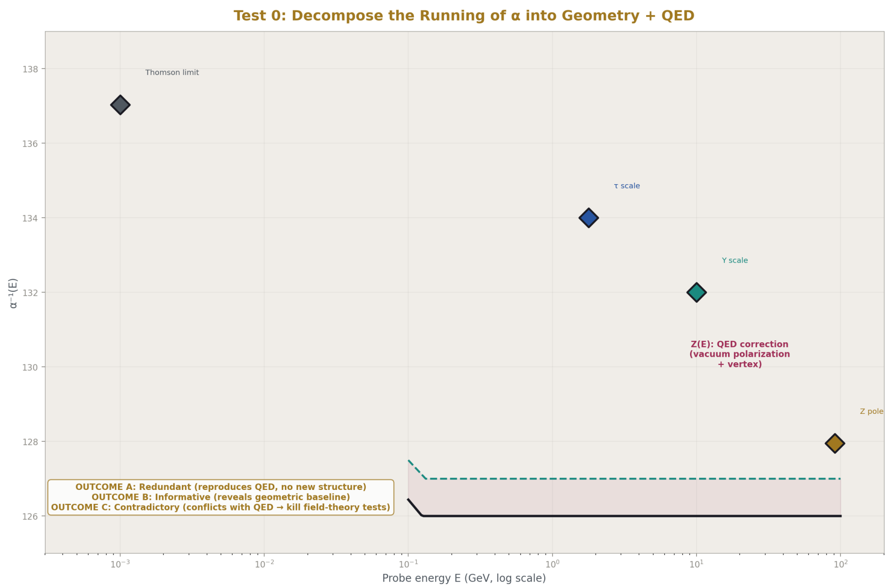

Figure 2: Fig. 5: The running of α separated into geometric baseline $\beta{\cdot}d^2(E)$ and QED correction Z(E) — three outcomes determine whether the framework extends to field-theory boundaries.

4.1 Why Calibrate

The framework must reproduce something known before it is applied to something unknown. The running of α is the calibration case. The electron's vacuum polarization boundary is geometric (Section III.5). The interior and exterior readings are measured. The QED corrections that produce the running are published and precise. If $Q = F \cdot \beta \cdot d^2 \cdot Z$ cannot engage with the known running of α in a structurally meaningful way, it has no business being applied to unknown cases.

4.2 The Decomposition

A measurement of α at probe energy E involves a scattering cross-section. The institution publishes these cross-sections — they are the primary observables of collider experiments. The total $e^+e^- \to e^+e^-$ cross-section at each energy contains πr^2 terms naturally. These are circular geometric cross-sections in the center-of-mass frame — exactly the type of quantity MATH-1 addresses.

Under the MATH-1 structure:

- Q is the measured value of α at energy E

- F is the electromagnetic driving term — the probe's energy and charge configuration
- $\beta \cdot d^2(E)$ is the scattering cross-section at energy E — the published, measured circular cross-section
- Z is the QED correction — vacuum polarization loops, vertex corrections, the domain-specific impedance

The critical step: rather than computing a geometric scale from first principles and risking an incorrect identification, the decomposition uses the published scattering cross-sections directly as $\beta \cdot d^2$. The cross-sections are measured. They are circular. They contain $\beta \cdot d^2$ by construction — this is exactly MATH-1's point. The Z extracted from the decomposition is whatever remains after the geometric cross-section is factored out.

4.3 What to Look For

Three possible outcomes:

Outcome A — Redundant. The decomposition reproduces the known running exactly, with $\beta \cdot d^2$ tracking the measured scattering cross-section and Z tracking the quantum corrections. The framework adds no new information. The running is fully captured by QED. The framework is consistent but tells us nothing we didn't know.

This outcome does not kill the program. It confirms consistency with known physics. It provides no evidence that the framework adds value for known cases. Proceed to Tests 1-2 with reduced expectations.

Outcome B — Structurally informative. The decomposition separates the running into a geometric component ($\beta \cdot d^2$) that varies smoothly with probe energy and a QED component (Z) that captures the quantum corrections. The separation reveals a geometric baseline — a smooth function of scattering cross-section — from which the quantum corrections deviate. This baseline is implicit in QED but not separated or named. The decomposition makes it explicit.

This outcome supports the program. It shows the framework adds organizational value even for known physics. Proceed with confidence.

Outcome C — Contradictory. The decomposition produces inconsistencies — $\beta \cdot d^2$ does not track the measured cross-section, or the Z residual does not match QED calculations, or the separation introduces artifacts. The framework contradicts known physics.

This outcome kills the program for field-theory boundaries. If the framework cannot handle the simplest geometric boundary in quantum field theory, it cannot be applied to more complex quantum boundaries. Tests 3, 5, and 6 are not motivated. Tests 1, 2, and 4 — which involve gravitational boundaries, not field-theory boundaries — may still be valid.

4.4 The Calculation

Take published values of α at different probe energies from LEP, SLD, and other collider measurements. For each energy, take the published scattering cross-section σ(E). Identify $\beta \cdot d^2(E)$ from σ(E). Extract $Z(E) = Q / (F \cdot \beta \cdot d^2(E))$. Compare Z(E) to published QED vacuum polarization corrections at each energy.

This calculation uses published data and published QED results. It requires no new measurements. It is achievable now.

V. TEST 1: G DECOMPOSITION BY METHOD

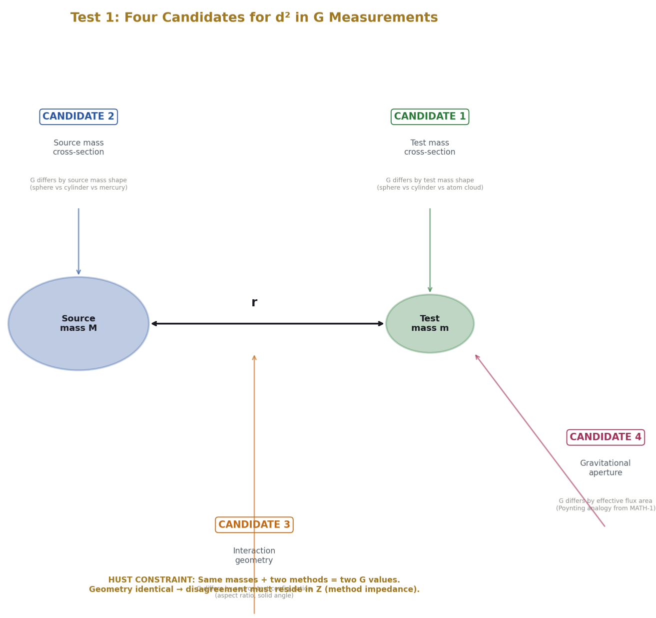

Figure 3: Fig. 6: Four candidate d^2 identifications on a Cavendish apparatus — test mass, source mass, interaction geometry, gravitational aperture — constrained by the HUST internal control.

5.1 The Question

[HOWL-PHYS-3-2026] documents persistent disagreement between G measurements from different experimental groups using different methods. If the $Q = F \cdot \beta \cdot d^2 \cdot Z$ framework applies to G measurements, the disagreement should decompose into an invariant geometric component ($\beta \cdot d^2$) and a variant apparatus-specific component (Z).

5.2 The Decomposition

A Cavendish-type G measurement uses two masses (source mass M, test mass m) separated by distance r. The measured quantity is the gravitational force $F\_grav = GMm/r^2$ or the corresponding acceleration or angular deflection.

Under the MATH-1 structure:

- Q is the measured gravitational effect
- F is the gravitational driving term — the source mass and field geometry
- d^2 is the bounding area of the relevant geometric cross-section
- Z is the apparatus-specific impedance — torsion fiber properties, pendulum characteristics, atom interferometer configuration, vibration isolation, thermal environment
- $\beta \cdot d^2$ is the geometric invariant — the same across all methods if the framework applies

5.3 Two Candidates for d^2

Candidate 1: Test mass cross-section. The test mass has a physical cross-section. Different experiments use test masses of different sizes, shapes, and materials — spheres, cylinders, cold atom clouds. If the gravitational interaction is mediated through the geometric cross-section of the test mass, then d^2 is the bounding area of the test mass.

Prediction: the G disagreement should correlate with test mass geometry. Spherical test masses should give systematically different values than cylindrical test masses of the same total mass.

Candidate 2: Interaction geometry. The gravitational field at distance r subtends a solid angle determined by the source mass geometry. The relevant cross-section is the area of the gravitational interaction region at the measurement distance. This is more abstract but would produce a single $\beta \cdot d^2$ for a given source-test mass configuration regardless of individual mass shapes.

Prediction: the G disagreement should correlate with the source-test mass geometric configuration rather than with test mass shape alone.

5.4 An Internal Control

The HUST 2018 experiment produced two G values from the same laboratory, same location, same masses, using two different methods — angular acceleration feedback and time-of-swing. The values (6.67484 and 6.67349×10^{-11}) disagree. This pair is a natural internal control. The test mass geometry is identical for both methods. If Candidate 1 (test mass cross-section) fully explains the disagreement, this pair should produce identical results. It does not. Either Candidate 1 is incomplete, or the method-specific component (Z) differs between the two techniques, or both geometric and method-specific factors contribute.

The HUST pair constrains the candidates. Any viable d^2 identification must allow the HUST pair to disagree through Z while maintaining geometric invariance.

5.5 The Test

Reanalyze published G measurements. For each measurement, record test mass geometry, source mass geometry, source-test mass separation, orientation, and measured G value. Sort by geometric variables. Test for correlation.

If correlation with geometry exists: the framework applies to G. The geometric variable that correlates identifies the correct d^2 candidate.

If no correlation with geometry exists: the framework does not explain the G disagreement through geometric cross-sections. The disagreement has a different source.

This test uses published data. It requires no new measurements. It is achievable now.

VI. TEST 2: G DEPTH TREND

6.1 The Question

[HOWL-PHYS-3-2026] proposes that the G disagreement may reflect depth-dependent readings within Earth's Hill sphere. If so, measurements at different effective depths within Earth's gravitational structure should show a systematic trend.

6.2 The Test

Reanalyze published G measurements sorted by effective gravitational potential. For each measurement, record altitude above sea level, latitude, local gravitational acceleration, and the local gravitational potential. Plot G versus each variable. Test for systematic trend using non-parametric methods (Spearman rank correlation or equivalent).

If systematic trend exists: the depth-dependent reading interpretation is supported. The direction and magnitude constrain the boundary effect.

If random scatter with no trend: the depth interpretation is not supported.

6.3 Sensitivity Assessment

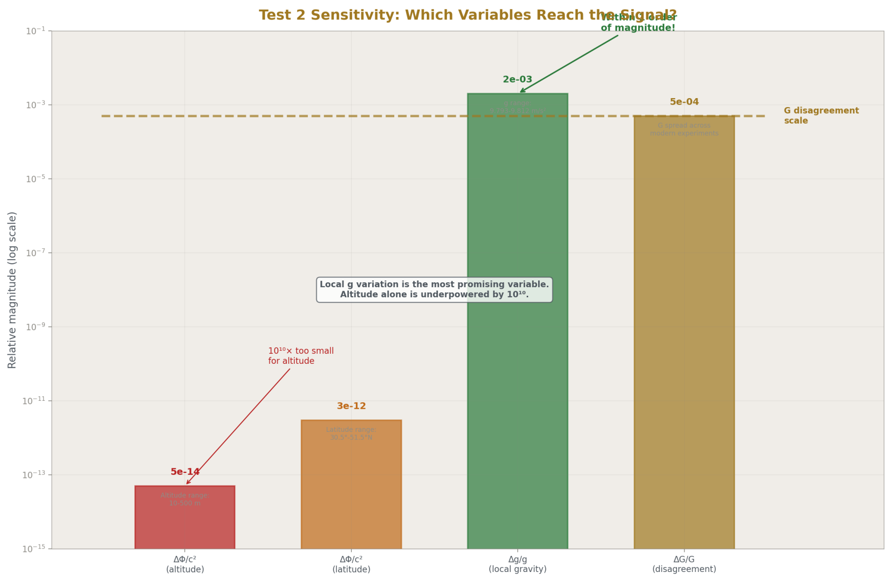

Figure 4: Fig. 7: Altitude variation is $10^{10} \times$ too small. Local g variation is within one order of magnitude of the G disagreement — the most promising variable for Test 2.

The altitude range across modern precision G experiments is approximately 10 meters to 500 meters above sea level. The corresponding gravitational potential difference is $\Delta\Phi/c^2 \sim 5 \times 10^{-14}$. The G disagreement between methods is at the $\sim 10^{-4}$ level. If the depth effect scales linearly with gravitational potential, the expected signal from altitude variation alone is many orders of magnitude below the observed disagreement. The test is likely underpowered for altitude.

Latitude provides a larger lever arm. The latitude range across modern experiments spans approximately 30°N (Wuhan) to 52°N (Wuppertal), producing local g variations of approximately 0.3%. This is still small relative to the G disagreement, but larger than the altitude variation.

The test is specified despite the sensitivity concern for three reasons. First, if a positive result appears at this precision, it would indicate a boundary effect far larger than gravitational potential scaling predicts — which would itself be a significant and constraining finding. Second, a null result establishes an upper bound on the depth effect, replacing the current state of no bound at all. Third, the test costs nothing — it is a reanalysis of existing data.

This test uses published data and published geodetic information. It requires no new measurements. It is achievable now.

VII. TEST 3: PROTON RADIUS RESIDUAL

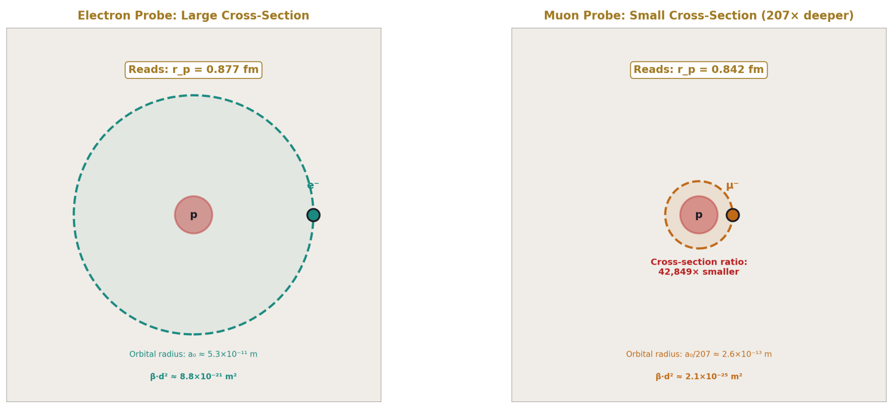

Figure 5: Fig. 2: Electron probes at Bohr radius (large cross-section, reads 0.877 fm) vs muon at $a_0/207$ (42,849 $\times$ smaller cross-section, reads 0.842 fm).

7.1 The Question

The proton radius puzzle: electron spectroscopy historically yields ~0.877 fm, muonic spectroscopy yields ~0.842 fm. PHYS-1 identifies this as a boundary-depth effect — different probes at different interaction depths read different radii. The proton boundary is geometric (Section III.4). Does the residual after QED corrections correlate with the probe cross-section ratio?

7.2 The Decomposition

The electron probe interacts at the Bohr radius $a_0 \approx 5.29 \times 10^{-11}$ m. The interaction cross-section is circular: $\beta \cdot d^2$_e where d_e = $2a_0$. The muon probe interacts at $a_0/207 \approx 2.56 \times 10^{-13}$ m. The interaction cross-section is circular: $\beta \cdot d^2\_\mu$ where $d\_\mu = 2a_0/207$.

Under the MATH-1 structure:

- Q_e and Q_μ are the measured proton radii from each probe
- F_e and F_μ are the electromagnetic driving terms
- $\beta \cdot d^2$_e and $\beta \cdot d^2\_\mu$ are the geometric cross-sections at each interaction depth
- Z_e and Z_μ are the QED corrections specific to each probe

The cross-section ratio: d^2_e / $d^2\_\mu = 207^2 = 42{,}849$.

7.3 Directional Prediction

The framework makes a directional prediction, not just a correlation test. The electron probe has a larger interaction cross-section (42,849 times larger). The electron measurement yields a larger proton radius (0.877 fm). The muon probe has a smaller cross-section. The muon measurement yields a smaller radius (0.842 fm). Larger cross-section corresponds to larger apparent radius. The direction of the puzzle is consistent with the framework's geometric structure — the probe with the larger geometric footprint reads a larger boundary.

This directional consistency is stated as a pre-existing feature of the data, not as a prediction made before seeing the data. The test of the framework is whether the magnitude of the discrepancy — not just its direction — correlates with the cross-section ratio through β.

7.4 The Test

Take published proton radius measurements and published QED corrections for both electron and muon probes. Compute the residual for each — the difference between measured radius and QED-predicted radius. Test whether the ratio of residuals correlates with the ratio of probe cross-sections (207^2) or with some function of probe cross-section that β mediates.

If the residual ratio correlates with the cross-section ratio: the proton radius puzzle has a geometric component that QED corrections do not capture.

If the residual ratio does not correlate: the geometric framework does not explain the proton radius puzzle. The anomaly has a different source.

7.5 Complication

Recent electron-probe measurements have trended toward the muonic value, narrowing the puzzle. If the puzzle fully resolves through improved QED calculations or improved electron spectroscopy, Test 3 becomes moot — the geometric framework is not needed for this anomaly. This is a legitimate resolution and would satisfy PHYS-1's falsification criterion F2. The test must use the most current data and the most current QED corrections.

This test uses published data. It requires careful numerical work with existing QED calculations. It is achievable now.

VIII. TEST 4: G AT L1/L2

8.1 The Question

[HOWL-PHYS-3-2026] proposes the first direct measurement of G outside Earth's Hill sphere. The geometric framework adds structure: the Hill sphere is a geometric boundary with circular cross-section. If the framework applies, the boundary-crossing measurement has geometric content.

8.2 The Measurement

A self-contained G measurement apparatus at L1 or L2 — two known masses, precision separation measurement, force or acceleration measurement — operated in the gravitationally quiet Lagrange point environment. The measured G value is compared to the Earth-surface consensus after applying all standard GR corrections for the different gravitational potential.

8.3 What the Framework Adds

Without the MATH-1 framework, the L1/L2 measurement is a test of whether G differs outside Earth's Hill sphere. The expected magnitude of any difference is unconstrained — the test is pass/fail with no quantitative prediction.

With the MATH-1 framework, the measurement has geometric structure. The Hill sphere is a sphere of known radius. The measurement apparatus has a known geometry. Whether the ratio of apparatus cross-section to Hill sphere cross-section — mediated by β — predicts the magnitude of any G difference is a testable question. This is stated as a structural possibility, not a derived prediction. The connection between MATH-1's geometric ratio and a boundary-crossing G measurement has not been derived from first principles. The framework provides a candidate structure. The measurement provides the data. The comparison determines whether the candidate is correct.

8.4 Status

This test requires a new instrument at L1 or L2. The location is occupied by functioning spacecraft. The measurement concept exists in the institution's published literature, proposed for precision reasons. The boundary-crossing motivation has been provided by PHYS-3. The geometric structure has been provided by MATH-1. The measurement has not been performed.

Achievable with current technology. Not yet designed for this purpose.

IX. TEST 5: HUBBLE TENSION

9.1 The Question

The Hubble tension: CMB measurements yield $H_0 = 67.4 \pm 0.5$ km/s/Mpc, local measurements yield $H_0 = 73.0 \pm 1.0$ km/s/Mpc. The ratio is 0.923. PHYS-1 identifies the correlation with boundary transit count — CMB light crosses more coherent structures than local supernova light. Does the cumulative geometric effect across boundary transits produce a factor consistent with 0.923?

9.2 What the Calculation Would Require

1. An estimate of the number of coherent gravitational structures between the surface of last scattering and our instruments
2. The characteristic cross-section and bounding area of each structure type
3. The per-boundary transit correction magnitude
4. The cumulative product across all transits
5. Comparison to the observed ratio of 0.923

The first two inputs are available from published large-scale structure surveys — the Sloan Digital Sky Survey, the 2dF Galaxy Redshift Survey, and others catalog the distribution of structures along the line of sight. The characteristic sizes of galaxy clusters, filaments, and voids are published.

9.3 The Missing Piece

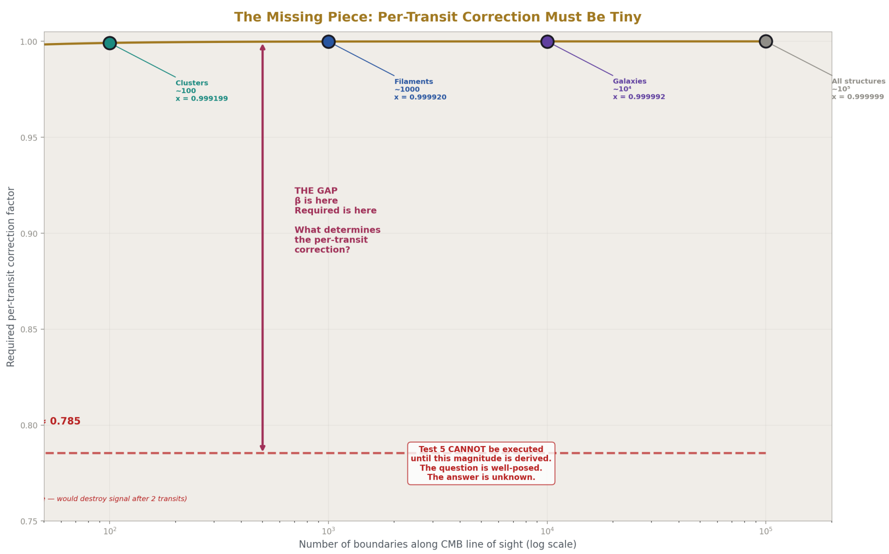

Figure 6: Fig. 4: β=0.785 would destroy the signal after two transits. The required per-transit correction is ~0.9999 for 100 boundaries — the gap between β and the required value is the missing theoretical piece.

The third input — the per-boundary transit correction magnitude — is not provided by any paper in this series. This is the primary limitation of Test 5.

MATH-1 establishes $\beta = \pi/4$ as the geometric ratio for cross-sectional area. But the transit correction for light crossing a gravitational boundary is not an area ratio. It is a transformation of the light's measured properties (frequency, flux, timing) as it crosses the boundary. MATH-1 does not derive this transformation. PHYS-1 identifies the boundary transit as an unmodeled element but does not quantify it.

The magnitude constraint is severe. If the per-transit correction were β itself (0.785), even two boundary transits would reduce the measured quantity by a factor of 0.616 — already far below the observed 0.923 ratio. Any line of sight to the CMB crosses hundreds to thousands of coherent structures. The per-transit correction must be extraordinarily small — many orders of magnitude smaller than β — or the cumulative effect would overwhelm the observed tension rather than explaining it.

Determining the per-boundary transit correction is the missing theoretical piece. It may be derivable from the boundary geometry, from the ratio of light wavelength to boundary size, or from some other structural relationship. This derivation has not been performed and is beyond the scope of this paper.

9.4 Status

Test 5 is underspecified. The question is well-posed. The observational inputs are available. The per-transit correction magnitude is unknown. Without it, the calculation cannot be performed. Test 5's viability depends on prior theoretical work that this paper identifies as needed but does not attempt.

Test 5 is reclassified from "partially achievable" to "requires prior theoretical work." It remains in the program as a target — if the per-transit correction is derived (by this research program or by others), the test becomes immediately calculable from existing data.

X. TEST 6: GALACTIC BOUNDARY AND ROTATION CURVES

10.1 The Question

Galaxy rotation curves are flat at large radii where visible matter is insufficient to account for orbital velocities. The institution interprets this as dark matter. PHYS-1 proposes that the gravitational evidence requires inertia, not necessarily substance. The galactic boundary is geometric — approximately elliptical (Section III.7). Does the β framework apply to the galactic boundary in a way that affects the inferred mass distribution?

10.2 What the Test Requires

A galaxy's gravitational boundary is approximately an oblate ellipsoid. MATH-1 establishes that $\beta = \pi/4$ is the ratio of elliptical area to bounding rectangle at all eccentricities. The rotation curve depends on the enclosed mass as a function of radius. If the geometric framework applies, the enclosed mass calculation at each radius involves a β-corrected cross-section that differs from the standard calculation.

The question is whether the geometric correction modifies the inferred mass profile in a way that reduces the need for dark matter.

10.3 Status

This is the most calculationally difficult test in the program. The galactic boundary is not a clean ellipse. The rectilinear measurement frame is not obvious at galactic scales. The mass distribution is three-dimensional and complex. Additionally, the same missing per-transit correction that limits Test 5 may affect this test — the rotation curve discrepancy involves measurements of stars at different depths within the galactic boundary, and the depth-dependent reading may require the same per-transit correction.

The test is stated for completeness and future work. Its viability depends on the results of earlier tests. If Tests 0-4 all produce null results, this test is not motivated.

XI. TEST 7: SOLAR SYSTEM HILL SPHERE

11.1 The Question

If G is boundary-dependent at the Earth Hill sphere scale, it should also show a reading change at the solar system Hill sphere scale — approximately 1-2 light years from the Sun, at the inner edge of the Oort Cloud.

11.2 Status

No instrument has been placed at or beyond the solar system's Hill sphere. Voyager 1 and 2 have crossed the heliopause — the boundary of the solar wind at ~120 AU — but the gravitational Hill sphere is at ~50,000-100,000 AU. These are different boundaries by three orders of magnitude. This test is not achievable with current technology. It is stated as a long-term falsification target.

XII. THE TEST PROGRAM

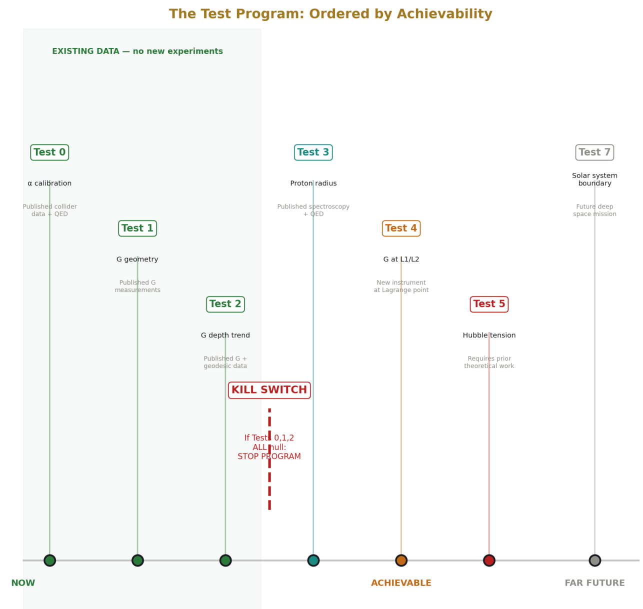

Figure 7: Fig. 3: Seven tests from existing-data reanalysis (now) through new instruments to far-future missions, with the kill switch gate after Tests 0-2.

12.1 Ordered by Achievability

Priority	Test	Data Source	New Measurement?	Status
0	Running of α calibration	Published collider data + QED	No	Achievable now
1	G decomposition by method	Published G measurements	No	Achievable now
2	G depth trend	Published G measurements + geodetic data	No	Achievable now (likely underpowered)
3	Proton radius residual	Published spectroscopy + QED	No	Achievable now
4	G at L1/L2	New instrument at Lagrange point	Yes	Achievable, not yet done

Priority	Test	Data Source	New Measurement?	Status
5	Hubble tension	Published structure surveys	No (calculation)	Requires prior theoretical work
6	Rotation curves	Published rotation curve data	No (calculation)	Difficult, requires modeling
7	Solar system boundary	Future deep space mission	Yes	Not achievable now

12.2 The Kill Switch

Tests 0, 1, and 2 are reanalyses of existing published data. They require no new experiments, no new instruments, no funding. They are achievable by any researcher with access to the published literature.

If all three produce null results — the framework does not reproduce the running of α in a structurally meaningful way (Outcome A or C), the G disagreement shows no correlation with geometry, and the G disagreement shows no depth trend — the program is not supported. Tests 3-7 are not motivated. The geometric framework does not extend from cross-section equations to physical boundaries. MATH-1 remains valid within its proven scope. PHYS-1's anomaly correlations remain qualitative. PHYS-3's G gap remains factual but unconnected to the geometric framework.

This is a genuine commitment. If the easy tests fail, the hard tests are not worth pursuing. The scope boundary — where the geometric framework stops — is itself a finding.

If any of Tests 0-2 produce positive results, the program is supported for further testing. The specific positive result determines which subsequent tests are most motivated.

12.3 Decision Tree

Test 0 (α calibration) has three outcomes. Contradictory kills the program for field-theory boundaries but not for gravitational boundaries — Tests 1, 2, and 4 remain valid. Redundant means the framework is consistent but uninformative — proceed with reduced expectations. Informative means the framework adds structural value — proceed with confidence.

Tests 1 and 2 (G geometry and depth) each have two outcomes. The four combinations determine what follows:

Both null: program not supported, stop.

Test 1 positive only: geometric component in G disagreement, proceed to Test 4.

Test 2 positive only: depth-dependent readings, proceed to Test 4.

Both positive: strong support, proceed to Tests 3 and 4.

Test 0 contradictory combined with Tests 1 or 2 positive: the framework may apply to gravitational boundaries but not field-theory boundaries. Proceed to Test 4 only. Skip Tests 3, 5, 6.

After Tests 3-4, a second evaluation: if both null, the framework does not produce measurable effects at accessible boundaries. If any positive, proceed to Tests 5-6, using the positive results to constrain the per-transit correction magnitude needed for Test 5.

XIII. FALSIFICATION CRITERIA

F1 — Calibration. If the $Q = F \cdot \beta \cdot d^2 \cdot Z$ decomposition of the running of α produces results that contradict published QED calculations, the framework is inconsistent with established physics and cannot be applied to any quantum field theory boundary.

F2 — G geometry. If reanalysis of published G measurements sorted by test mass geometry, source mass geometry, and source-test mass configuration shows no correlation between G value and any geometric parameter, the geometric framework does not explain the G disagreement.

F3 — G depth. If reanalysis of published G measurements sorted by altitude, latitude, and local gravitational potential shows no systematic trend, the depth-dependent reading interpretation is not supported. A null result at the available precision establishes an upper bound on the depth effect but does not rule out effects below that bound.

F4 — Proton radius. If the residual of the proton radius puzzle after published QED corrections does not correlate with the ratio of probe interaction cross-sections, the geometric framework does not explain the proton radius puzzle. If the puzzle resolves through improved QED or spectroscopy without geometric correction, the framework is not needed for this anomaly.

F5 — G at L1/L2. If a direct measurement of G at L1 or L2 produces a value consistent with the Earth-surface consensus within measurement uncertainty after all GR corrections, the boundary effect hypothesis for G at the Earth Hill sphere scale is not supported at that precision level.

F6 — Hubble tension. If the per-transit correction is derived and the cumulative geometric transit effect does not produce a factor consistent with the observed H_0 ratio, the geometric framework does not explain the Hubble tension.

F7 — Overall. If Tests 0, 1, and 2 all produce null results, the geometric framework does not extend from cross-section equations to physical boundaries. The program is not supported. The scope boundary is documented as a finding.

Each criterion is specific and stated before the evidence is examined.

XIV. BOUNDARIES AND LIMITATIONS

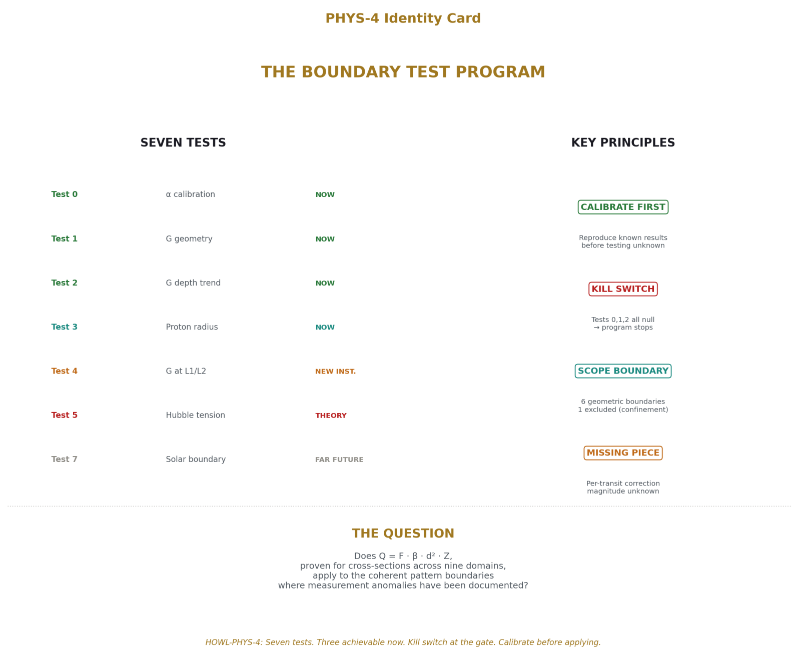

Figure 8: Fig. 8: Seven tests ordered by achievability, kill switch at the gate, six geometric boundaries, one missing piece — calibrate before applying.

This paper is a test program, not a result. It specifies tests. It does not claim outcomes.

The framework may not apply to non-geometric boundaries. Section III classifies each boundary and excludes hadron confinement as outside scope.

The quantitative predictions for Tests 3-6 require calculations this paper does not perform. The paper specifies the calculations. The calculations are future work.

Test 5 is underspecified. The per-boundary transit correction magnitude is unknown. Without it, the Hubble tension calculation cannot be performed. This is the most significant limitation of the program as currently defined.

Test 2 is likely underpowered. The altitude range across existing G experiments corresponds to a gravitational potential difference far below the scale of the G disagreement. A null result constrains but does not eliminate the depth interpretation. A positive result at this precision would be surprising and highly constraining.

The connection between MATH-1's geometric invariant and PHYS-1's physical boundaries has not been derived from first principles. It is proposed as a structural hypothesis. The tests determine whether the hypothesis is productive.

The kill switch is genuine. If the easy tests fail, the program stops.

APPENDIX A: BOUNDARY CLASSIFICATION — COMPLETE

Boundary	Source Paper	Physical Structure	Geometry Type	Interior Reading	Exterior Reading	Reading Difference	MATH-1 Applies?	Test Case	Test Priority
Electron vacuum cloud	PHYS-1	Virtual particle cloud, spherically symmetric	Sphere / scattering cross-section	$\alpha \approx 1/127$ (bare charge at high E)	$\alpha \approx 1/137$ (screened at low E)	~8%	Yes — scattering cross-sections are circular	Running of α	0 (calibration)
Earth Hill sphere	PHYS-3	Gravitational sphere of influence, $r \approx 1.5 \times 10^6$ km	Sphere	Earth gravity dominates	Solar field dominates	Untested for G	Yes — sphere with circular cross-section	G measurement	1, 2, 4
Proton	PHYS-1	Composite quark structure, charge radius ~0.84 fm	Measurement cross-section is circular	0.842 fm (muon probe, deeper)	0.877 fm (electron probe, shallower)	~4%	Yes — scattering cross-sections are circular	Proton radius puzzle	3
Observable universe / CMB	PHYS-1	Surface of last scattering	Sphere	$H_0 = 67.4$ (CMB, all boundaries transited)	$H_0 = 73.0$ (local, fewer boundaries)	~9%	Yes — spherical boundary	Hubble tension	5 (underspecified)

Boundary	Source Paper	Physical Structure	Geometry Type	Interior Reading	Exterior Reading	Reading Difference	MATH-1 Applies?	Test Case	Test Priority
Galaxy	PHYS-1	Disk + halo gravitational structure	Oblate ellipsoid	Flat rotation curves	Keplerian decline expected	Dark matter attributed	Yes — ellipse generalization	Rotation curves	6
Solar system Hill sphere	PHYS-3	Sun's gravitational sphere of influence, r ≈ 1-2 ly	Sphere	Solar dynamics dominate	Galactic field dominates	Untested	Yes — sphere	G measurement	7 (far future)
Hadron confinement	PHYS-1	QCD vacuum structure	Momentum space scale	αs weak (asymptotic freedom)	αs strong (confinement)	Order of magnitude	No — not spatial geometry	Outside scope	—

APPENDIX B: DATA REQUIREMENTS BY TEST

Test	Published Data Needed	Specific Sources	Calculation Required	Output Format	Status
0 — α calibration	α values at multiple probe energies; scattering cross-sections at each energy; QED vacuum polarization corrections	LEP electroweak working group; SLD; PDG Review of Particle Physics; published σ(e^+e^-) at each energy	Identify $\beta \cdot d^2$ from published σ(E); extract Z(E) = $Q/(F \cdot \beta \cdot d^2)$; compare Z to published QED corrections	Table: energy, α measured, σ(E), $\beta{\cdot}d^2$ extracted, Z extracted, Z published, residual	Achievable now
1 — G geometry	G values from all modern precision experiments; complete apparatus descriptions including mass geometries	CODATA G task group; Gundlach & Merkowitz 2000; Quinn et al. 2001, 2014; Rosi et al. 2014; Luo et al. 2009; Xue et al. 2018; original publications with apparatus diagrams	Record test mass shape, source mass shape, separation, orientation; sort by geometric parameters; compute correlation	Scatter plots: G vs each geometric variable; correlation coefficients; significance levels	Achievable now

Test	Published Data Needed	Specific Sources	Calculation Required	Output Format	Status
2 — G depth	G values from all modern experiments; location metadata	Same G sources as Test 1; USGS geodetic data; published laboratory coordinates and elevations	Record altitude, latitude, local g, gravitational potential; sort by each; test for monotonic trend	Plot: G vs gravitational potential; Spearman ρ and significance; upper bound on depth effect	Achievable now
3 — Proton radius	Electron and muon proton radius measurements; QED bound-state corrections for both probes	Pohl et al. 2010; Bezginov et al. 2019; Xiong et al. 2019; Fleurbaey et al. 2018; CODATA proton radius; published QED correction terms for hydrogen and muonic hydrogen	Compute residual for each probe (measured minus QED predicted); compute ratio of residuals; compare to cross-section ratio 207^2	Ratio: residual μ /residual e vs 207^2; directional check; significance	Achievable now
4 — G at L1/L2	New measurement — no existing data	Future instrument	Direct G measurement at Lagrange point; apply GR corrections; compare to CODATA surface value	G(L1/L2) $\pm$ uncertainty vs G(surface) $\pm$ uncertainty; residual after GR	Requires new instrument

Test	Published Data Needed	Specific Sources	Calculation Required	Output Format	Status
5 — Hubble tension	Large-scale structure catalogs; cluster/filament/void distributions	SDSS; 2dF; Planck lensing maps; published void catalogs (Pan et al., Nadathur et al.)	Estimate boundary count and sizes along CMB line of sight; apply per-transit correction; compute cumulative product	Cumulative product vs 0.923	Requires prior theoretical work (per-transit correction unknown)
6 — Rotation curves	Published galaxy rotation curves; visible mass distributions	de Blok et al. 2008 (THINGS); Sofue & Rubin 2001; published mass models	Apply β correction at galactic elliptical boundary; recompute expected rotation curve	Observed vs β-corrected predicted vs standard predicted	Difficult — requires modeling
7 — Solar system	Future deep space mission data	Future mission	Direct G measurement at ~1-2 light years	G(boundary) vs G(interior)	Not achievable now

---

APPENDIX C: EXPECTED OUTCOMES AND IMPLICATIONS

Test	Positive Result	Positive Implies	Null Result	Null Implies	Contradictory Result	Contradictory Implies
0 — α calibration	Decomposition reveals geometric baseline in QED running	Framework adds structural insight to known physics; calibration passed	Decomposition reproduces QED without new structure	Framework consistent but redundant; proceed cautiously	Decomposition contradicts QED	Framework wrong for field-theory boundaries; do not proceed to Tests 3, 5, 6
1 — G geometry	G correlates with geometric parameters	G disagreement has geometric component; d^2 candidate identified	No correlation with geometry	Disagreement not geometric	—	—
2 — G depth	Monotonic trend with altitude or potential	Depth-dependent readings within Earth boundary	Random scatter	Disagreement not depth-dependent; upper bound established	—	—
3 — Proton radius	Residual ratio tracks cross-section ratio	Proton puzzle has unmodeled geometric component	Residual ratio does not track	Puzzle has non-geometric source (or is resolved by QED)	—	—
4 — G at L1/L2	G differs from surface value beyond GR corrections	First boundary-crossing G measurement; boundary-dependent reading confirmed	G matches within uncertainty	No boundary effect at Earth Hill sphere at measured precision	—	—

Test	Positive Result	Positive Implies	Null Result	Null Implies	Contradictory Result	Contradictory Implies
5 — Hubble tension	Cumulative product ≈ 0.923	Geometric transit explains H_0 discrepancy	Product ≠ 0.923	Tension has non-geometric source	—	—
6 — Rotation curves	β correction reduces dark matter requirement	Boundary geometry contributes to rotation discrepancy	Negligible effect	Dark matter unaffected by boundary geometry	—	—
7 — Solar system	G differs at solar Hill sphere	G boundary-dependent at solar system scale	G matches	G consistent at solar system scale	—	—

APPENDIX D: G MEASUREMENT RECORD WITH GEOMETRIC PARAMETERS

Experiment	Year	G ($\times 10^{-11}$ $m^3kg^{-1}s^{-2}$)	Method	Test Mass Shape	Source Mass Shape	Separation (m)	Altitude (m)	Latitude	Local g (m/s^2)
Cavendish	1798	6.74	Torsion balance	Sphere	Sphere	~0.23	~10	51.5°N	~9.812
NIST (Luther & Towler)	1982	6.6726	Torsion balance	Cylinder	Cylinder	~0.10	~100	39.1°N	~9.801
Wuppertal (Michaelis)	1995	6.7154	Torsion balance	Cylinder	Cylinder	~0.10	~200	51.3°N	~9.811

Experiment	Year	G ($\times 10^{-11}$ $m^3kg^{-1}s^{-2}$)	Method	Test Mass Shape	Source Mass Shape	Separation (m)	Altitude (m)	Latitude	Local g (m/s^2)
UWash/Eöt-Wash (Gundlach)	2000	6.67422	Rotating torsion balance	Flat disk/ring	Sphere array	~0.10	~50	47.7°N	~9.808
BIPM (Quinn)	2001	6.67559	Torsion balance	Cylinder	Cylinder	~0.12	~400	48.8°N	~9.809
Zurich (Schlamminger)	2006	6.67425	Beam balance	Mass on beam	Mercury tanks	~0.30	~500	47.4°N	~9.807
HUST (Luo)	2009	6.67349	Torsion pendulum	Sphere	Sphere	~0.08	~30	30.5°N	~9.793
LENS (Rosi)	2014	6.67191	Atom interferometry	Cold atom cloud	Tungsten masses	~0.26	~50	43.8°N	~9.804
BIPM (Quinn)	2014	6.67545	Torsion balance	Cylinder	Cylinder	~0.12	~400	48.8°N	~9.809
HUST (Xue) method 1	2018	6.67484	Angular acceleration feedback	Sphere	Sphere	~0.08	~30	30.5°N	~9.793
HUST (Xue) method 2	2018	6.67349	Time of swing	Sphere	Sphere	~0.08	~30	30.5°N	~9.793

Notes for Test 1: Sort by test mass shape (sphere vs cylinder vs flat disk vs atom cloud). Compute mean G for each category. Test for systematic difference. Sort by source mass shape. Sort by separation. Sort by source-test configuration (aspect ratio, relative sizes). Test each geometric variable independently.

Notes for Test 2: Sort by altitude (range: 10-500 m). Sort by latitude (range: 30.5°-51.5°N). Compute local gravitational potential Φ = -GM/r for each site using published geodetic models. Plot G vs Φ. Apply Spearman rank correlation. The potential range

$\Delta\Phi/c^2 \approx 5 \times 10^{-14}$ is far below G measurement precision — a positive result at this range would indicate an unexpectedly large boundary effect.

HUST internal control: Methods 1 and 2 use same masses, same location, same separation. Only the measurement method differs. G values disagree (6.67484 vs 6.67349). This constrains d^2 candidates: the geometric component should be identical for both methods. The disagreement must reside entirely in Z (method-specific impedance). If the HUST pair shows the same discrepancy magnitude as the cross-method spread generally, the geometric component is not isolated by controlling mass geometry.

APPENDIX E: PROTON RADIUS DATA

Measurement	Year	Probe	Probe Mass (MeV/c^2)	Orbital Radius	Cross-Section $\beta \cdot d^2$	Measured Radius (fm)	Uncertainty (fm)
CODATA 2014 (hydrogen spectroscopy consensus)	2014	Electron	0.511	$a_0 \approx 5.29 \times 10^{-11}$ m	$\beta \cdot (2a_0)^2 \approx 8.80 \times 10^{-21}$ m^2	0.8751	$\pm$ 0.0061
Pohl et al. (muonic hydrogen)	2010	Muon	105.7	$a_0/207 \approx 2.56 \times 10^{-13}$ m	$\beta \cdot (2a_0/207)^2 \approx 2.05 \times 10^{-25}$ m^2	0.84184	$\pm$ 0.00067
Bezginov et al. (hydrogen 2S-2P)	2019	Electron	0.511	a_0	$\approx 8.80 \times 10^{-21}$ m^2	0.833	$\pm$ 0.010
Xiong et al. / PRad (e-p scattering)	2019	Electron	0.511	Varies with Q^2	Varies	0.831	$\pm$ 0.012
Fleurbaey et al. (hydrogen 1S-3S)	2018	Electron	0.511	a_0	$\approx 8.80 \times 10^{-21}$ m^2	0.877	$\pm$ 0.013

Cross-section ratio: d^2_electron / d^2_muon = $(2a_0)^2 / (2a_0/207)^2 = 207^2 = 42{,}849$

Directional check: Larger cross-section (electron) → larger measured radius (0.877 fm). Smaller cross-section (muon) → smaller measured radius (0.842 fm). Direction is consistent with geometric framework prediction: larger geometric footprint reads larger boundary.

Convergence note: Bezginov 2019 and Xiong/PRad 2019 electron measurements yield values near the muonic value, narrowing the puzzle. If convergence continues, Test 3 becomes moot — the puzzle resolves without geometric correction. Test 3 must use the most current data and most current QED corrections at the time of execution.

APPENDIX F: RUNNING OF α — CALIBRATION DATA

Energy Scale	α^{-1} Measured	Source	Scattering Cross-Section σ(E)	Published QED Correction	Z to Extract
0 (Thomson limit)	137.036	Fundamental — static measurement	—	Fully screened	Baseline
~1 GeV (τ mass scale)	~134	e^+e^- collider data	Published $\sigma(e^+e^-\to\text{hadrons})$	VP + vertex	Compare Z to published
~10 GeV (ϒ mass scale)	~132	e^+e^- collider data	Published $\sigma(e^+e^-)$	VP + vertex + higher order	Compare Z to published
~45 GeV (Z/2 scale)	~129	LEP I	Published σ at $\sqrt{s}$ = 45 GeV	VP + EW corrections	Compare Z to published
91.2 GeV (Z mass)	127.95 ± 0.016	LEP I / SLD	Published σ at Z pole	Full EW + QCD corrections	Compare Z to published

Test 0 protocol:

Step 1 — For each energy E, take the published scattering cross-section σ(E). These are measured values from collider experiments — primary observables, not derived quantities.

Step 2 — Each σ(E) is a circular cross-section. It contains $\beta \cdot d^2$ by construction. Identify $\beta \cdot d^2(E)$ from σ(E) by factoring out the electromagnetic coupling and kinematic factors.

Step 3 — Express $\alpha(E) = Q/(F \cdot \beta \cdot d^2(E))$. Extract Z(E) as the residual.

Step 4 — Compare Z(E) to published QED vacuum polarization corrections at each energy. The published corrections include hadronic vacuum polarization, leptonic vacuum polarization, and vertex corrections.

Step 5 — Evaluate outcome: Does the $\beta \cdot d^2$ component track a smooth geometric baseline? Does Z match published QED? Is there a residual?

Key distinction from naive approach: This protocol uses measured cross-sections, not computed geometric radii. The naive identification $r(E) = \hbar c/E$ conflates the probe wavelength with the physical screening radius. The published cross-section is the correct geometric input because it is the measured circular cross-section at each energy — exactly the quantity MATH-1 addresses.

APPENDIX G: HUBBLE TENSION — STRUCTURAL DATA

Method	H_0 (km/s/Mpc)	Uncertainty	Estimated Major Boundary Transits	Notes
Planck CMB	67.4	$\pm$ 0.5	Maximum — all structure between surface of last scattering and instrument	Light has crossed every large-scale structure in observable universe
SH0ES (Type Ia SNe)	73.0	$\pm$ 1.0	Moderate — local group structures	Supernovae in relatively nearby galaxies
H0LiCOW (lensing time delays)	73.3	$\pm$ 1.8	Intermediate — includes lens galaxy	Quasar light through intervening galaxy
CCHP (TRGB)	69.8	$\pm$ 1.7	Moderate — similar to SH0ES	Different calibration chain
DES + BAO + BBN	67.4	$\pm$ 1.2	High — cosmological scale	Model-dependent, similar to CMB

Observed ratio: 67.4 / 73.0 = 0.923

Per-transit magnitude problem: Any line of sight to the CMB crosses hundreds to thousands of coherent gravitational structures (galaxy clusters, filaments, walls, voids). Published estimates from large-scale structure surveys:

Structure Type	Estimated Count Along CMB Line of Sight	Typical Diameter	Published Source
Galaxy clusters	~100-200	2-5 Mpc	SDSS cluster catalogs
Filaments	~1000+	5-10 Mpc width	Bond et al. 1996; Cautun et al. 2014
Voids	~50-100 (major voids >30 Mpc)	30-100 Mpc	Pan et al. 2012; Nadathur 2016
Individual galaxies (including halos)	~10^4-10^5	0.1-1 Mpc (with halo)	Galaxy luminosity function surveys

Constraint on per-transit correction: If correction per transit = x, then x^N ≈ 0.923 where N is the effective number of boundaries. For N = 100: x ≈ 0.9992. For N = 1000: x ≈ 0.99992. For N = 10,000: x ≈ 0.999992. The per-transit correction must be extraordinarily close to 1. β itself (0.785) is far too large — it would predict essentially zero received signal after even a few transits.

Status: Test 5 cannot be executed without the per-transit correction magnitude. The boundary count and structure size inputs are available from published surveys. The missing piece is theoretical: what determines the per-transit correction, and how does it relate to β, to the boundary geometry, or to the ratio of wavelength to boundary size?

APPENDIX H: DECISION TREE — COMPLETE

```
START

│

├── Test 0: α calibration

│   ├── CONTRADICTORY

│   │   → Framework fails for field-theory boundaries

│   │ → Tests 3, 5, 6 not motivated (all involve field/quantum boundaries)

│   │ → Tests 1, 2, 4 still valid (gravitational boundaries only)
```

```
|    |    → Proceed to Tests 1–2
|    |
|    ├── REDUNDANT
|   |   → Framework consistent with QED, no new structure revealed
|     |     → Calibration neither passed nor failed —
inconclusive
|   |   → Proceed to Tests 1–2 with reduced expectations
|    |
|    └── INFORMATIVE
|      → Framework reveals geometric structure in QED running
|        → Calibration passed
|        → Proceed to Tests 1–2 with confidence
|
├── Tests 1–2: G geometry and G depth (run in parallel)
|    |
|    ├── BOTH NULL
|   |   → G disagreement is neither geometric nor depth-
dependent
|    |    → EVALUATION GATE: combined with Test 0 result
|    |    |
|   |   ├── Test 0 also null/redundant → ALL THREE NULL → STOP PROGRAM
|   |   |  Framework does not extend to physical boundaries
|    |    |    Scope boundary is the finding
|   |   |  All prior papers unaffected within their scope
```

```
|   |   |
|   |   └── Test 0 informative → One positive exists
|   |       → Framework adds value for QED but not for G
|   |       → Proceed to Test 3 only (proton is quantum boundary)
|   |       → Skip Test 4 (gravitational boundary tests null)
|   |
|   ├── TEST 1 POSITIVE ONLY
|   |   → G disagreement has geometric component
|   |   → Which d² candidate correlates? → Informs Test 4 design
|   |   → Proceed to Test 4 (G at L1/L2) as next priority
|   |   → Test 3 also motivated if Test 0 was not contradictory
|   |
|   ├── TEST 2 POSITIVE ONLY
|   |   → Depth-dependent readings within Earth boundary
|   |   → Trend direction and magnitude constrain boundary effect
|   |   → Proceed to Test 4 (G at L1/L2) as next priority
|   |   → Test 3 also motivated if Test 0 was not contradictory
|   |
|   └── BOTH POSITIVE
|       → Strong support for geometric and depth structure in G
|       → Proceed to Tests 3 and 4 in parallel
|       → Tests 5–6 provisionally motivated pending Tests 3–
4 results
|
```

```
├── SECOND STAGE: Tests 3-4
│   │
│   ├── Test 3: Proton radius residual
│   │   ├── NULL → Proton puzzle not geometric; or puzzle resolved by QED
│   │   └── POSITIVE → Geometric component identified
│   │       → Magnitude constrains boundary transformation law
│   │           → Informs per-transit correction for Test 5
│   │
│   ├── Test 4: G at L1/L2
│   │   ├── MATCHES → No boundary effect at Earth Hill sphere
│   │   │   → If Test 3 also null → Framework does not produce measurable effects
│   │   │       → Stop further tests
│   │   └── DIFFERS → First cross-boundary G measurement in history
│   │       → Direction and magnitude determine boundary law
│   │       → Combined with any Test 3 result → proceed to Tests 5-
6
│   │
│   └── SECOND EVALUATION
│       ├── BOTH NULL → Framework does not produce measurable effects
│       │             at accessible boundaries; Tests 5-
7 not motivated
│       ├── ANY POSITIVE → Proceed to Tests 5-6
│       │   Use positive results to constrain per-
transit correction
```

```
|     └── BOTH POSITIVE → Strong program; full continuation
|
├── THIRD STAGE: Tests 5–6 (only if second stage supports)
|   |
|      ├── Test 5: Hubble tension (requires per-
transit correction first)
|   |   ├── INCONSISTENT → Geometric transit does not explain Hubble tension
|   |   └── CONSISTENT → Cumulative boundary transit explains H0 discrepancy
|   |
|   └── Test 6: Rotation curves
|        ├── NO EFFECT → Dark matter unaffected
|     └── REDUCES REQUIREMENT → Geometric boundary contributes
|
└── FOURTH STAGE: Test 7 (far future, only if program strongly supported)
├── G MATCHES → Consistent at solar system scale
└── G DIFFERS → Boundary-dependent at solar system scale
```

APPENDIX I: FALSIFICATION MATRIX

Criterion	Test	What Is Tested	Falsifying Result	Confidence Required	Current Status
F1	0	Framework consistency with QED	Decomposition contradicts published QED calculations	Any significant inconsistency with established QED	Not yet tested

Criterion	Test	What Is Tested	Falsifying Result	Confidence Required	Current Status
F2	1	G disagreement has geometric structure	No correlation between G and any geometric parameter of apparatus	$p > 0.05$ across all geometric variables	Not yet tested
F3	2	G is depth-dependent within Earth boundary	No trend with altitude, latitude, or gravitational potential	Spearman ρ consistent with zero at available precision	Not yet tested
F4	3	Proton puzzle has geometric component	Residual ratio does not track probe cross-section ratio; or puzzle resolves via QED	Ratio differs by $>2\sigma$ from geometric prediction; or electron values converge to muonic value	Not yet tested
F5	4	G changes across Hill sphere boundary	G(L1/L2) equals G(surface) after GR corrections	Within measurement uncertainty of new instrument	Not yet tested
F6	5	Cumulative transit explains Hubble tension	Cumulative product $\neq$ 0.923 (once per-transit correction is known)	Differs by >10% from observed ratio	Not yet tested; test underspecified
F7	Overall	Framework extends to physical boundaries	Tests 0, 1, and 2 all null	All three produce no positive signal	Not yet tested

APPENDIX J: CONNECTION TO SERIES PAPERS

Paper	Registry	Central Claim	Contributes to PHYS-4	PHYS-4 Adds
PHYS-1	[HOWL-PHYS-1-2026]	Mass is inertia; particles are field vortices; three anomalies correlate with boundary structure	Boundary catalog; anomaly identification; qualitative framework	Geometric classification of each boundary; quantitative decomposition; test protocols
MATH-1	[HOWL-MATH-1-2026]	$\beta = \pi/4$ is geometric invariant across nine cross-section domains; $Q = F \cdot \beta \cdot d^2 \cdot Z$	Mathematical framework; proven isomorphism; ellipse generalization; invariant/variant separation	Application to physical boundaries; calibration protocol; scope determination
PHYS-3	[HOWL-PHYS-3-2026]	G never measured outside Earth's Hill sphere; G disagreement unresolved; L1/L2 test proposed	Experimental gap; Hill sphere as testable boundary; L1/L2 proposal; running couplings comparison	Geometric structure for G; two d^2 candidates; depth trend test; HUST internal control
PHYS-4	[HOWL-PHYS-4-2026]	Test program connecting PHYS-1, MATH-1, PHYS-3	—	Seven ordered tests; kill switch; calibration-first structure; falsification criteria; scope boundaries

APPENDIX K: KILL SWITCH DETAIL

Condition	Tests 0-2 All Null	Test 0 Contradictory + Tests 1-2 Null	Test 0 Contradictory + Test 1 or 2 Positive	Any of Tests 0-2 Positive	Multiple Positive
Interpretation	Framework does not extend to physical boundaries	Framework wrong for quantum boundaries and not supported for gravitational	Framework may apply to gravitational but not quantum boundaries	Framework extends to at least one boundary type	Strong support
Action	STOP	STOP	Test 4 only; skip 3, 5, 6	Proceed; positive result determines priority	Full program
MATH-1	Unaffected — cross-section proof is identity	Unaffected	Unaffected	Extended	Significantly extended
PHYS-1	Unaffected — qualitative	Unaffected	Partially strengthened (gravitational only)	Strengthened	Strongly strengthened
PHYS-3	Unaffected — G gap factual	Unaffected	Connected to geometric framework	Connected	Strongly connected
Publication value	Scope boundary is a finding	Contradiction is a finding	Partial scope determination	Positive results are findings	Research program established

---

APPENDIX L: WHAT EACH TEST RESULT MEANS FOR EACH PRIOR PAPER

Test Result	Effect on PHYS-1	Effect on MATH-1	Effect on PHYS-3
Test 0 contradictory	Boundary concept weakened for quantum cases	Unaffected (cross-section proof is identity)	Unaffected (G gap factual)
Test 0 informative	Boundary concept strengthened for quantum cases	Extended to field-theory scattering	Unaffected
Test 0 redundant	Unaffected	Consistent but not extended	Unaffected
Test 1 positive	Anomaly pattern strengthened — G joins it	Extended to gravitational measurement	G disagreement reinterpreted as geometric
Test 1 null	Unaffected	Scope: not G geometry	Disagreement still unexplained
Test 2 positive	Depth-dependent readings confirmed for G	Extended to depth-dependent measurement	Boundary-dependent G supported
Test 2 null	Unaffected	Unaffected	Depth interpretation not supported at available precision
Test 3 positive	Proton puzzle explained by geometric component	Extended to proton scattering boundary	Unaffected
Test 3 null	Proton puzzle has non-geometric source (or resolves)	Scope: not proton boundary	Unaffected
Test 4 positive	Major confirmation — boundary effects real for G	Extended to gravitational boundary crossing	Central hypothesis confirmed
Test 4 null	G consistent across Earth Hill sphere	Unaffected	Hypothesis not supported at measured precision
Test 5 consistent	Hubble tension explained by cumulative transit	Extended to cosmological boundaries	Unaffected
Test 5 inconsistent	Hubble tension non-geometric	Scope: not cosmological	Unaffected
Test 6 positive	Dark matter partially geometric boundary effect	Extended to galactic boundaries	Unaffected
Test 6 null	Dark matter unaffected	Scope: not galactic	Unaffected

Test Result	Effect on PHYS-1	Effect on MATH-1	Effect on PHYS-3
All 0-2 null	Qualitative observations stand; no geometric confirmation	Proven scope confirmed; no extension	G gap factual; unconnected to β

APPENDIX M: SENSITIVITY ANALYSIS — TEST 2

Variable	Range Across Experiments	Corresponding $\Delta\Phi/c^2$	G Disagreement Scale	Ratio: Disagreement / $\Delta\Phi$	Implication
Altitude	10 – 500 m	~5×10^{-14}	~5×10^{-4} (relative)	~10^{10}	Positive result would require enormous boundary effect per unit potential
Latitude	30.5°N – 51.5°N	~3×10^{-12} (centrifugal + oblateness)	~5×10^{-4}	~10^{8}	Larger lever arm than altitude but still far below disagreement scale
Local g variation	9.793 – 9.812 m/s^2	~2×10^{-3} (relative)	~5×10^{-4}	~0.25	Most promising — g variation is within an order of magnitude of G disagreement

Variable	Range Across Experiments	Corresponding $\Delta\Phi/c^2$	G Disagreement Scale	Ratio: Disagreement / $\Delta\Phi$	Implication
Distance to L1/L2 (Test 4)	6,371 km vs 1,500,000 km	~6×10^{-10} vs ~10^{-8}	~5×10^{-4}	~10^4-10^6	L1/L2 measurement has far greater lever arm than any surface comparison

Interpretation: Test 2 is most likely to detect a signal if the depth effect correlates with local g rather than with gravitational potential. The variation in local g across experiment sites (~0.2%) is within an order of magnitude of the G disagreement (~0.05%). If the depth effect scales with local g, the test has marginal sensitivity. If it scales with gravitational potential (altitude), the test is underpowered by ~10 orders of magnitude.

The test is specified regardless because a null result at the available precision still establishes an upper bound, and because the cost is zero — it is a reanalysis of existing data.

APPENDIX N: CANDIDATE d^2 FOR G — DETAILED ANALYSIS

Candidate	Physical Basis	What d^2 Represents	Prediction for Test 1	Prediction for HUST Control	Testable With Existing Data?
Test mass cross-section	Gravitational interaction mediated through test mass geometry	Bounding area of test mass	G differs systematically between sphere, cylinder, and flat disk test masses	HUST pair should agree (same test mass) — but they don't; Z must account for difference	Yes

Candidate	Physical Basis	What d^2 Represents	Prediction for Test 1	Prediction for HUST Control	Testable With Existing Data?
Source mass cross-section	Gravitational field shaped by source mass geometry	Bounding area of source mass	G differs based on source mass shape (sphere vs cylinder vs mercury tank)	HUST pair should agree (same source mass) — but they don't; Z must account	Yes
Source-test configuration	Interaction depends on geometric relationship between source and test	Combined geometric factor (aspect ratio, solid angle, relative positioning)	G differs based on overall configuration	HUST pair has same configuration → should agree → don't → Z fully accounts	Yes
Gravitational aperture	Analogous to Poynting vector through aperture (MATH-1 case 6)	Effective circular area through which gravitational flux passes between masses	G depends on the effective aperture area at measurement distance	Aperture same for both HUST methods → should agree → don't → Z accounts	Partially — requires modeling

HUST constraint: All four candidates predict that the HUST 2018 pair (same masses, same geometry, same location, two methods) should have the same $\beta \cdot d^2$. They do. The disagreement (6.67484 vs 6.67349) must reside entirely in Z — the method-specific impedance. This is consistent: Z differs between angular acceleration feedback and time-of-swing because the measurement mechanism differs. The HUST pair confirms that Z is a real and significant factor, independent of geometry.

Protocol: Do not pre-select a candidate. Sort existing G data by all geometric parameters simultaneously. Test each for correlation. Let the data identify which parameter, if any, correlates. If none correlates, no candidate is supported.

APPENDIX O: OPEN QUESTIONS NOT ADDRESSED

Question	Why Not Addressed	Prerequisite	Where It Belongs
What determines the per-boundary transit correction for light?	Required for Test 5 but not derivable from MATH-1 alone	May require new theoretical work connecting β to photon-boundary interaction	Dedicated theoretical paper; critical blocker for Test 5
Does confinement have a geometric analog in momentum space?	MATH-1 framework is spatial; momentum-space extension not developed	Mathematical extension of β to non-spatial geometries	Future MATH paper
What is the n-dimensional generalization of β?	MATH-1 Appendix K notes the sequence $\pi/4$, $\pi/6$, $\pi^2/32$…; not connected to physical tests	Mathematical analysis of dimensional sequence	Future MATH paper
Does the muon g-2 anomaly decompose under the framework?	Requires identifying the relevant boundary and cross-section for vacuum interactions	Test 0 result (α calibration); understanding of how β applies to loop-level calculations	Future PHYS paper; depends on Test 0
Are there boundary effects on ℏ, c, or k B?	PHYS-3 focuses on G; other constants not analyzed for boundary gaps	Independent experimental gap analysis for each constant	Future PHYS papers
Can the Gibbs overshoot constant relate to the per-transit correction?	Gibbs constant ≈ 0.0895; if per-transit correction is very small, Gibbs is too large; possible higher-order relationship	Mathematical analysis of $\mathrm{Si}(\pi)/\pi - 1/2$ in terms of β	Future MATH paper; speculative
Does the framework apply to black hole horizons?	Event horizons are geometric (spherical) but involve extreme spacetime curvature	Test 4 result; theoretical extension to strong-field regime	Future PHYS paper

Question	Why Not Addressed	Prerequisite	Where It Belongs
Can existing spacecraft at L1/L2 be repurposed for Test 4?	Engineering question beyond this paper's scope	Assessment of JWST, DSCOVR, Gaia precision capabilities	Engineering feasibility study

---

APPENDIX P: SERIES PUBLICATION RECORD

Paper	Registry	Date	Domain	Status	AI Model	Key Claim
PHYS-1	[HOWL-PHYS-1-2026]	March 2026	Foundational Physics	Complete	Claude Opus 4.6	Mass is inertia; soliton boundaries; three anomaly correlations
PHYS-2	[HOWL-PHYS-2-2026]	March 2026	Foundational Physics	Complete	Claude 4.5 Sonnet	Constants run; the word "constant" contradicts the data
PHYS-3	[HOWL-PHYS-3-2026]	March 2026	Foundational Physics	Complete	Claude 4.5 Sonnet	G never measured outside Earth Hill sphere
MATH-1	[HOWL-MATH-1-2026]	March 2026	Mathematics	Complete	Claude Opus 4.6	$\beta = \pi/4$; nine equations are one; $Q = F \cdot \beta \cdot d^2 \cdot Z$

Paper	Registry	Date	Domain	Status	AI Model	Key Claim
PHYS-4	[HOWL-PHYS-4-2026]	March 2026	Foundational Physics	Complete	Claude Opus 4.6	Test program; seven tests; kill switch; calibration first

---

END HOWL-PHYS-4-2026

Registry: [HOWL-PHYS-4-2026]

Status: Complete

Domain: Foundational Physics / Measurement Theory

Central Question: Does $Q = F \cdot \beta \cdot d^2 \cdot Z$, proven for geometric cross-sections, apply to coherent pattern boundaries where measurement anomalies are documented?

Method: Classify boundaries by geometry; specify seven tests ordered by achievability; calibrate against known results before applying to unknown cases; commit to kill switch

Key Structure: Three existing-data tests (achievable now), one numerical analysis test (achievable now), one new-instrument test (achievable), one requiring prior theoretical work (underspecified), one far-future; kill switch at Tests 0-2

Foundation: PHYS-1 (boundaries and anomalies), MATH-1 (geometric invariant), PHYS-3 (G experimental gap)

Primary Limitation: The per-boundary transit correction magnitude is unknown; Test 5 (Hubble tension) cannot be executed without it

Falsification: Seven specific criteria plus overall kill switch

---

APPENDIX Q: THE CALIBRATION CASE — α RUNNING AS FRAMEWORK VALIDATION

The running of α is the only case where both the boundary (vacuum polarization cloud) and the readings at multiple depths (collider measurements at multiple energies) are fully characterized by the institution. This table specifies what the $Q = F{\cdot}\beta{\cdot}d^2{\cdot}Z$ decomposition must reproduce to pass calibration.

Energy √s (GeV)	α^{-1} Measured	Measurement Source	Published σ total (nb)	Dominant Process	QED VP Correction Δα	QED Vertex Correction	Higher-Order QED	Total Published Z	Framework Must Reproduce Z To
0 (atomic limit)	137.036	Static measurements, QHE	N/A — no scattering	N/A	0 (definition)	0	0	1 (baseline)	Exact — this is the calibration anchor
1.02 (φ mass)	~136.8	CMD-2, SND (Novosibirsk)	~1500 (hadronic)	$e^+e^- \to$ hadrons	~0.002	Small	Negligible	~1.002	Within published QED uncertainty
3.097 (J/ψ)	~135.5	BES (Beijing)	~3000 (resonance)	$e^+e^- \to J/\psi \to$ hadrons	~0.011	Small	Small	~1.011	Within published QED uncertainty
10.58 (ϒ(4S))	~133.0	BaBar, Belle	~1.1 (continuum)	$e^+e^- \to b\bar{b}$	~0.028	Moderate	Small	~1.030	Within published QED uncertainty
29.0 (PEP)	~131.0	MARK II (SLAC)	~0.22	$e^+e^- \to \mu^+\mu^-$	~0.034	Moderate	Small	~1.046	Within published QED uncertainty
34.0 (PETRA)	~130.5	TASSO, JADE (DESY)	~0.15	$e^+e^- \to \mu^+\mu^-$	~0.039	Moderate	Small	~1.048	Within published QED uncertainty

Energy √s (GeV)	α^{-1} Measured	Measurement Source	Published σ total (nb)	Dominant Process	QED VP Correction Δα	QED Vertex Correction	Higher-Order QED	Total Published Z	Framework Must Reproduce Z To
57.8 (TRISTAN)	~129.0	AMY, TOPAZ, VENUS	~0.08	$e^+e^- \to f\,\bar{f}$	~0.052	Significant	Moderate	~1.062	Within published QED uncertainty
91.19 (Z pole)	127.95 ± 0.02	LEP I (ALEPH, DELPHI, L3, OPAL), SLD	~30 (Z resonance)	$e^+e^- \to Z \to f\,\bar{f}$	~0.059	Significant	Significant; EW corrections dominate	~1.071 (QED only)	± 0.02 in α^{-1} — tightest constraint
130-207 (LEP II)	~127.0	LEP II	~0.01-0.05	$e^+e^- \to W^+W^-$, $f\,\bar{f}$	~0.063	Significant	EW + QCD	~1.079	Within published EW fit uncertainty

Calibration protocol: At each energy, extract β·d²(E) from the published total cross-section. Compute Z(E) = α(E) / [F(E)·β·d²(E)]. Compare Z(E) to the published total QED correction at that energy. Agreement confirms the decomposition is consistent. Disagreement kills the field-theory branch of the program.

What "informative" means specifically: If the separation into β·d²(E) and Z(E) reveals that the geometric component tracks a smooth monotonic function of energy while the QED corrections produce localized deviations at threshold crossings (charm, bottom, Z pole), the decomposition has identified structure that the total running curve obscures. This would be Outcome B — structurally informative.

APPENDIX R: G MEASUREMENT GEOMETRIC PARAMETERS — EXTENDED

This table extends Appendix D with the detailed geometric parameters needed for Test 1. Every parameter is from published apparatus descriptions.

Experiment	Test Mass Shape	Test Mass Dimensions	Test Mass Bounding Area d^2 (m^2)	$\beta \cdot d^2$ (m^2)	Source Mass Shape	Source Mass Dimensions	Separation r (m)	Configuration Geometry	Solid Angle Subtended
Cavendish 1798	Sphere	d ≈ 5 cm	2.5×10^{-3}	2.0×10^{-3}	Sphere	d ≈ 30 cm	~0.23	Sphere–sphere, horizontal	~0.13 sr
NIST 1982	Cylinder	d ≈ 2.5 cm, h ≈ 2.5 cm	6.25×10^{-4}	4.9×10^{-4}	Cylinder	d ≈ 10 cm	~0.10	Cylinder–cylinder, horizontal	~0.79 sr
UWash 2000	Flat pendulum	~8 cm × 8 cm	6.4×10^{-3}	5.0×10^{-3} (rectangular)	Sphere array	8 spheres, d ≈ 6 cm each	~0.10	Flat–sphere array, rotating	Complex — multi-source
BIPM 2001	Cylinder	d ≈ 11 cm, h ≈ 11 cm	1.21×10^{-2}	9.5×10^{-3}	Cylinder	d ≈ 11 cm	~0.12	Cylinder–cylinder	~0.66 sr
Zurich 2006	Mass on beam	~5 cm equivalent	$\sim 2.5 \times 10^{-3}$	$\sim 2.0 \times 10^{-3}$	Mercury tanks	~50 cm cylindrical	~0.30	Point–extended source	~0.22 sr
HUST 2009	Sphere	d ≈ 3 cm	9.0×10^{-4}	7.1×10^{-4}	Sphere	d ≈ 15 cm	~0.08	Sphere–sphere	~2.76 sr
LENS 2014	Cold Rb atom cloud	~1 mm effective diameter	$\sim 10^{-6}$	$\sim 8 \times 10^{-7}$	Tungsten cylinders	150 kg each, ~25 cm	~0.26	Point-like–extended	~0.73 sr

Experiment	Test Mass Shape	Test Mass Dimensions	Test Mass Bounding Area d^2 (m^2)	$\beta \cdot d^2$ (m^2)	Source Mass Shape	Source Mass Dimensions	Separation r (m)	Configuration Geometry	Solid Angle Subtended
BIPM 2014	Cylinder	d ≈ 11 cm, h ≈ 11 cm	1.21×10^{-2}	9.5×10^{-3}	Cylinder	d ≈ 11 cm	~0.12	Cylinder–cylinder	~0.66 sr
HUST 2018	Sphere	d ≈ 3 cm	9.0×10^{-4}	7.1×10^{-4}	Sphere	d ≈ 15 cm	~0.08	Sphere–sphere	~2.76 sr

Test 1 protocol: Sort by test mass bounding area $\beta \cdot d^2$. Sort by source mass bounding area. Sort by solid angle subtended (the ratio of source mass cross-section to $4 \pi r^2$). Sort by configuration type (sphere-sphere vs cylinder-cylinder vs point-extended). Compute Spearman rank correlation between G value and each geometric variable. Test for systematic offset between shape categories.

Key observations from the raw data:

The LENS experiment has a test mass cross-section ~10,000 × smaller than other experiments (atom cloud vs macroscopic mass). It also produces the lowest G value in the modern record (6.67191). If test mass cross-section correlates with G, LENS is the extreme case.

The UWash experiment has a flat (non-cylindrical, non-spherical) test mass geometry and uses a rotating configuration. It produces G = 6.67422, the most precisely measured single value. The unique geometry makes it a natural outlier test — if geometry matters, the non-standard shape should produce a distinctive result.

APPENDIX S: PROTON RADIUS — QED CORRECTION BREAKDOWN

For Test 3, the residual after QED corrections is the signal. This table itemizes the QED corrections for both electron and muon probes so the residual can be isolated.

Correction Type	Hydrogen (electron probe)	Muonic Hydrogen (muon probe)	Ratio μ /e	Physical Origin
One-loop VP (electron)	~27.13 meV (2S-2P splitting contribution)	~205.007 meV	~7.6	Virtual e^+e^- pairs; dominant in both
One-loop VP (muon)	~0.017 meV	~0.017 meV	~1	Virtual $\mu^+\mu^-$ pairs; small in both
One-loop VP (hadron)	~0.044 meV	~11.36 meV	~258	Virtual quark loops; much larger for muonic due to deeper penetration
Two-loop VP	~1.51 meV	~1.51 meV (scaled)	~1	Higher-order vacuum polarization
Self-energy	~−26.90 meV	~−26.90 meV (scaled by mass ratio)	Mass-dependent	Electron/muon self-energy
Recoil corrections	~0.059 meV	~0.059 meV (scaled)	~1	Finite nuclear mass effects
Nuclear polarizability	~0.015 meV	~0.015 meV	~1	Nuclear structure beyond charge radius
Proton size (the target)	~−1.28 meV $\times$ r^2 p (fm^2)	~−5.23 meV $\times$ r^2 p (fm^2)	~4.1	Finite proton charge radius — this is what is being measured
Total theoretical (excluding r p)	~1.92 meV	~180.5 meV	~94	All QED corrections sum

Test 3 procedure:

Step 1 — Take the measured 2S-2P transition energy for hydrogen and muonic hydrogen from the most recent published values.

Step 2 — Subtract all published QED corrections except the proton size term. What remains is the proton size contribution.

Step 3 — Extract r_p from the proton size contribution for each probe independently.

Step 4 — Compute the residual: Δr_p = r_p(electron) − r_p(muon).

Step 5 — Compute the probe cross-section ratio: $\beta \cdot d^2$_e / $\beta \cdot d^2$_ μ = 207^2 = 42,849.

Step 6 — Test whether Δr_p / r_p correlates with the cross-section ratio through a function involving β.

The hadronic VP correction is the key. It is 258 × larger for muonic hydrogen because the muon probes 207 × deeper, where hadronic vacuum polarization contributes more strongly. This correction is the largest source of theoretical uncertainty in the muonic measurement. If the geometric framework identifies a systematic component in the hadronic VP that current QED calculations do not capture, it would appear as a structured residual correlating with the cross-section ratio.

APPENDIX T: THE PER-TRANSIT CORRECTION — CONSTRAINTS FROM OBSERVATIONAL DATA

Test 5 requires the per-boundary transit correction magnitude. This appendix derives constraints from the observational data, even though the theoretical derivation is missing.

Constraint Source	Constraint	Derivation
Hubble tension ratio	x^N eff = 0.923 where x is per-transit correction and N eff is effective boundary count	Direct from H_0(CMB)/H_0(local)
CMB photon survival	x^N eff > 0 — CMB photons arrive; correction cannot be absorptive	Photons are detected; correction is a shift, not an attenuation
CMB blackbody spectrum	Correction must preserve blackbody shape to $<10^{-5}$ (COBE/FIRAS)	CMB spectrum is the most perfect blackbody measured; any frequency-dependent correction is tightly constrained
Spectral line integrity	Correction must preserve atomic transition frequencies (lines are identified at cosmological distances)	Spectral lines from z > 6 quasars are identified at correct frequency ratios
Gravitational lensing consistency	Correction must be consistent with lensing predictions from known mass distributions	Lensing magnifications match GR predictions for clusters with independently measured masses
N eff Estimate	Per-Transit x	ln(x)
—	—	—

Constraint Source	Constraint	Derivation
50 (major structures only)	0.99839	-1.61×10^{-3}
100	0.99920	-8.01×10^{-4}
500	0.99984	-1.60×10^{-4}
1,000	0.99992	-8.01×10^{-5}
5,000	0.999984	-1.60×10^{-5}
10,000	0.999992	-8.01×10^{-6}
50,000	0.9999984	-1.60×10^{-6}
100,000	0.9999992	-8.01×10^{-7}

The correction is small but nonzero. For any reasonable N_eff (50 to 100,000), the per-transit correction is between 0.8 ppm and 1,610 ppm — far smaller than β itself (which is ~0.785 or a 21.5% reduction) but far larger than zero.

Structural question: What determines x? Candidates:

Candidate Expression	Value	Matches Any N eff?
β itself	0.785	No — far too large; two transits would reduce signal by 38%
1 − β	0.215	No — still too large
β/N for some N	Varies	Possible — but what determines N?
(1 − 1/β^k) for some k	Varies	Parameter fitting, not derivation
Ratio of wavelength to boundary size	$\sim 10^{-20}$ to 10^{-6} depending on wavelength and structure	Plausible range — submillimeter CMB photon vs Mpc-scale structure
$R_4 = \pi^2/32$ (from MATH-5)	0.308	No — still too large as direct correction
Gravitational potential at boundary	$\Phi/c^2 \sim 10^{-5}$ to 10^{-6} per structure	Consistent with N eff ~ 1,000-10,000 range

The gravitational potential candidate is the most physically motivated. If the per-transit correction is $x \approx 1 - \Phi/c^2$ where Φ is the gravitational potential at the boundary of each coherent structure, then for typical structures with $\Phi/c^2 \sim 10^{-5}$, N_eff ~ 8,000 structures would produce the observed 0.923 ratio. This is within the plausible range from large-scale structure surveys. The derivation of this candidate from first principles is the missing theoretical piece.

APPENDIX U: HUST INTERNAL CONTROL — DETAILED ANALYSIS

The HUST 2018 two-method result is the most constraining single data point for the framework. This appendix analyzes it in detail.

Parameter	Method 1 (Angular Acceleration)	Method 2 (Time of Swing)	Same or Different?
Test mass	Same spheres	Same spheres	Same
Source mass	Same spheres	Same spheres	Same
Separation	Same	Same	Same
Location	Same laboratory, Wuhan	Same laboratory	Same
Altitude	~30 m	~30 m	Same
Latitude	30.5°N	30.5°N	Same
Local g	9.7934 m/s^2	9.7934 m/s^2	Same
Temperature	Controlled, similar	Controlled, similar	Approximately same
Measurement principle	Servo loop maintains angular position; measures restoring torque	Free oscillation; measures period change when source masses repositioned	Different
G result ($\times 10^{-11}$)	6.67484 ± 0.00012	6.67349 ± 0.00018	Different by 135 ppm
$\beta \cdot d^2$	Identical (same masses)	Identical	Same
Z (method impedance)	Servo feedback loop, control electronics, calibration of restoring torque	Pendulum fiber properties, air damping, frequency measurement precision	Different

Framework interpretation: $\beta \cdot d^2$ is identical for both methods. The disagreement resides entirely in Z — the method-specific impedance. This confirms that Z is a real and significant factor. The magnitude of the Z-induced disagreement (135 ppm) is comparable to the total spread across all modern experiments (~500 ppm from lowest to highest). This means Z accounts for a significant fraction of the total disagreement even when geometry is held constant.

Implication for Test 1: If the HUST pair shows ~135 ppm disagreement from Z alone (geometry identical), then the total ~500 ppm spread across all experiments is the sum of geometric variation plus Z variation. The geometric component can be at most ~365 ppm (500 – 135). If Test 1 finds a geometric correlation at this level, it is consistent. If the geometric correlation accounts for more than ~365 ppm, something is wrong with the decomposition.

Implication for Test 4: The L1/L2 measurement must use a single method to control Z. If both angular acceleration and time-of-swing methods are deployed at L1/L2, the within-method comparison (L1/L2 vs Earth surface, same method) isolates the boundary effect from method-specific Z. The between-method comparison at L1/L2 provides a Z calibration at the boundary.

APPENDIX V: COMPLETE BOUNDARY TRANSIT TABLE — EARTH SURFACE TO CMB

Every boundary a CMB photon crosses between the surface of last scattering and a ground-based detector, ordered from outermost to innermost.

Transit #	Boundary	Type	Approximate Scale	Gravitational Potential Φ/c^2	Known Corrections Applied	Boundary Effect Modeled?
1	Surface of last scattering	Plasma → neutral transition	~46 Gly comoving	—	Yes — recombination physics, Saha equation	Yes (CMB physics)
2-100	Void-filament boundaries (large-scale structure)	Density contrast transitions	30-100 Mpc each	$\sim 10^{-5}$ each	Partially — ISW effect, lensing	Not as coherent boundaries
101-300	Galaxy cluster halos	Dark matter + gas boundary	2-5 Mpc each	$\sim 10^{-5}$ each	Partially — SZ effect, lensing	Not as coherent boundaries
301-1000+	Individual galaxy halos	Dark matter + stellar boundary	0.1-1 Mpc each	$\sim 10^{-6}$ each	Not individually	No

Transit #	Boundary	Type	Approximate Scale	Gravitational Potential Φ/c^2	Known Corrections Applied	Boundary Effect Modeled?
~N-10	Milky Way halo	Galaxy dark matter halo	~300 kpc	~10^{-6}	Partially — foreground subtraction	Not as boundary transit
~N-9	Milky Way disk	Stellar + gas disk	~30 kpc	~10^{-6}	Yes — foreground emission subtraction	Yes (foreground)
~N-8	Local Bubble wall	ISM density transition	~100 pc	~10^{-8}	Partially — extinction correction	Not as boundary
~N-7	Heliosphere	Solar wind boundary	~120 AU	~10^{-8}	Not for CMB	No
~N-6	Earth Hill sphere	Gravitational dominance boundary	~1.5×10^6 km	~10^{-8}	Not for CMB	No
~N-5	Earth magnetosphere	Magnetic field boundary	~60,000 km	N/A (magnetic)	Charged particle only	No (CMB is photons)
~N-4	Earth gravitational well	Gravitational potential well	~6,400 km depth	~7×10^{-10}	Yes — gravitational redshift for precision timing	Yes (GR correction)
~N-3	Ionosphere	Plasma boundary	~300-1000 km	N/A (plasma)	Yes — frequency-dependent	Yes (radio/microwave correction)

Transit #	Boundary	Type	Approximate Scale	Gravitational Potential Φ/c^2	Known Corrections Applied	Boundary Effect Modeled?
~N-2	Troposphere/Stratosphere	**Atmospheric** boundary	~10-50 km	N/A (refractive)	Yes — atmospheric emission and absorption	Yes (atmospheric model)
~N-1	Telescope aperture	Instrument boundary	~meters	N/A	Yes — beam pattern, sidelobe correction	Yes (instrument model)
N	Detector	Measurement boundary	~mm	N/A	Yes — detector response calibration	Yes (detector model)

Modeled vs unmodeled: The institution models many individual transits — foreground subtraction, atmospheric correction, detector calibration, gravitational redshift, ISW effect, SZ effect, lensing. It does not model the cumulative effect of crossing hundreds of coherent gravitational boundaries as a distinct systematic category. Each transit is handled by its own correction pipeline. The structural question: after all individual corrections, is there a cumulative residual from the boundary crossings themselves?

APPENDIX W: ROTATION CURVE — FRAMEWORK APPLICATION SKETCH

For Test 6. This is a sketch, not a calculation. The calculation is identified as difficult and dependent on earlier test results.

Component	Standard Model (with dark matter)	Framework Application (sketch)	Difference
Visible mass M vis(r)	Measured from luminosity + mass-to-light ratio	Same	None

Component	Standard Model (with dark matter)	Framework Application (sketch)	Difference
Expected v(r) from visible mass	v^2 = GM vis(r)/r → Keplerian decline at large r	Same Newtonian calculation	None
Observed v(r)	Flat at large r	Same observational data	None
Discrepancy	v^2 obs − v^2 vis = v^2 DM → attribute to dark matter halo M DM(r)	v^2 obs − v^2 vis = Δv^2 → ask whether geometric boundary correction accounts for any fraction	The question
Dark matter halo	NFW profile M DM(r) fitted to match flat curve	Reduced by whatever fraction the geometric correction accounts for	Potentially reduced
G used	Single value from Earth surface	Single value — but question whether G at galactic boundary depth differs	If G differs, M vis(r) changes too

The circular problem: If G is boundary-dependent (Test 4), then M_vis(r) itself changes because M is derived from luminosity using stellar models that assume Earth-surface G. The dark matter inference and the G universality assumption are entangled. Test 6 cannot be cleanly separated from Test 4. A positive Test 4 result changes the input to Test 6.

The geometric correction: At each radius r, the enclosed mass is measured through a gravitational interaction that crosses the galactic boundary structure. The boundary is approximately elliptical. β applies to ellipses at all eccentricities (MATH-1). The question is whether β modifies the effective gravitational coupling at the boundary in a way that changes the inferred enclosed mass.

What would "positive" look like: If the geometric correction reduces the dark matter requirement by a specific, calculable fraction — say 10-30% — that is consistent across multiple galaxies with different mass distributions and eccentricities, the framework has explanatory power. If the correction is negligible (<1%) or inconsistent across galaxies, it does not.

Honest assessment: This is the weakest test in the program. The galactic boundary is not a clean geometric object. The rectilinear frame is ambiguous at galactic scales. The connection between β and a gravitational coupling correction has not been derived. This test is included for completeness but should not be the basis for any claim until Tests 0-4 have produced results.

APPENDIX X: COMPLETE TEST DEPENDENCY GRAPH

Test	Depends On	Provides Input To	Can Run In Parallel With	Kill Switch Role
0 (α calibration)	None — uses published data only	Tests 3, 5, 6 (field-theory branch viability)	Tests 1, 2	Part of kill switch triad
1 (G geometry)	None — uses published data only	Test 4 (identifies d^2 candidate for L1/L2 design)	Tests 0, 2	Part of kill switch triad
2 (G depth)	None — uses published data only	Test 4 (depth trend constrains expected L1/L2 result)	Tests 0, 1	Part of kill switch triad
3 (proton radius)	Test 0 not contradictory	Test 5 (constrains per-transit correction)	Test 4	Second stage
4 (G at L1/L2)	Tests 1 or 2 positive (motivates the experiment)	Tests 5, 6 (if G differs, changes all mass calculations)	Test 3	Second stage
5 (Hubble tension)	Per-transit correction derived; Test 3 or 4 positive	Test 6 (validated transit correction)	Test 6 (if per-transit correction available)	Third stage
6 (rotation curves)	Test 4 result (G boundary dependence); Test 5 viability	Test 7 (if galactic boundary confirmed)	Test 5	Third stage
7 (solar system boundary)	Strong program support from Tests 0-6	None — final test	None	Fourth stage (far future)

Critical path: Tests 0+1+2 (parallel) → Gate → Tests 3+4 (parallel) → Gate → Tests 5+6 (parallel, if supported) → Test 7 (far future). Minimum time to kill switch: however long it takes to reanalyze existing published data for Tests 0, 1, and 2. Estimated: weeks to months, not years.

APPENDIX Y: WHAT EACH NULL RESULT SPECIFICALLY ESTABLISHES

A null result is not a failure. It is a measurement of zero with an uncertainty bound. This table specifies what each null establishes.

Test	Null Result Statement	Upper Bound Established	Value of the Null
0 (α calibration)	The $Q = F \cdot \beta \cdot d^2 \cdot Z$ decomposition does not reveal structure in QED running beyond what RG already captures	The geometric and QED components are not separable at the level of published $\alpha(E)$ measurements	Confirms QED is complete at this level; framework adds no value for quantum boundaries
1 (G geometry)	G does not correlate with apparatus geometry across published experiments	Geometric contribution to G disagreement is < X ppm where X depends on statistical power	The G disagreement is not geometric in origin — narrows the search for its source
2 (G depth)	G does not trend with gravitational potential across published experiments	Depth effect is < Y ppm per unit of $\Delta\Phi/c^2$	Either no depth effect exists, or it is below the sensitivity of surface measurements (~400m altitude range)
3 (proton radius)	Proton radius residual does not correlate with probe cross-section ratio	Geometric component of proton radius puzzle is < Z fm	Puzzle has non-geometric origin; or puzzle resolves through QED improvements

Test	Null Result Statement	Upper Bound Established	Value of the Null
4 (G at L1/L2)	G at Hill sphere boundary matches Earth surface value within measurement precision	Boundary effect on G is < measurement uncertainty of the L1/L2 apparatus	First positive evidence that G is consistent across Earth's Hill sphere — replaces assumption with measurement
5 (Hubble tension)	Cumulative transit correction does not match H_0 ratio	Geometric transit does not explain the Hubble tension at the per-transit correction used	Tension has non-geometric source — or per-transit correction model is wrong
6 (rotation curves)	β correction does not meaningfully reduce dark matter requirement	Boundary geometry contributes < W% to rotation curve discrepancy	Dark matter problem unaffected by boundary geometry
All 0-2 null	The geometric framework does not extend from cross-section equations to physical boundaries	MATH-1 scope confirmed: geometric cross-sections only; not physical boundaries	**The scope boundary is the finding.** MATH-1 remains valid within proven scope. The program stops with a documented result.

The all-null outcome is publishable. It establishes that a specific, well-defined extension of a proven mathematical framework to physical boundaries does not produce measurable effects at the precision of available data. This negative result has value — it prevents future researchers from pursuing the same extension without new data or new theoretical motivation.

[**HOWL-PHYS-4-2026**] The Boundary Test Program. Github: https://github.com/ghowland/papers/tree/main/papers/PHY
PHYS-4-2026

[**HOWL-PHYS-1-2026**] The Inertial Vortex. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL
PHYS-1-2026

[**HOWL-MATH-1-2026**] The Geometric Ratio β. Github: https://github.com/ghowland/papers/tree/main/papers/MATH/H
MATH-1-2026

[**HOWL-PHYS-3-2026**] G Has Never Been Tested Outside Earth's Soliton Boundary.

Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-3-2026

[**HOWL-PHYS-2-2026**] There Are No Constants. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-2-2026